Electronic Devices

Classroom Workbook

ELECTRONICS AND COMM. ENGINEERING

GATE/ESE



Electronic Devices

Classroom Workbook

For ECE

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Sorting HAT Technologies Pvt. Ltd.

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GATE Syllabus

ECE

Energy bands in intrinsic and extrinsic semiconductors, equilibrium carrier concentration, direct and indirect band-gap semiconductors.

Carrier Transport: diffusion current, drift current, mobility and resistivity, generation and recombination of carriers, Poisson and continuity equations.

P-N junction, Zener diode, BJT, MOS capacitor, MOSFET, LED, photo diode and solar cell.

ESE Syllabus

E & T

Basics of semiconductors; Diode/Transistor basics and characteristics; Diodes for different uses; Junction & Field Effect Transistors (BJTs, JFETs, MOSFETs); Transistor amplifiers of different types, oscillators and

other circuits; Basics of Integrated Circuits (ICs); Bipolar, MOS and CMOS ICs; Basics of linear ICs, operational amplifiers and their applications linear/non-linear; Optical sources/detectors; Basics of Opto electronics and its applications.

Contents

Sr.	Chapters
1.	Semiconductor Physics
2.	P-N Junction Diode
3.	Basics of BJT
4.	MOSFETs
5.	Special Purpose Diodes

■■■■■

1 Semiconductor Physics

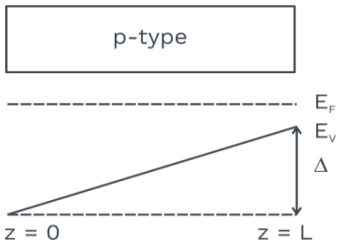
MCQ, MSQ and NAT

Q.1 A bar of silicon is doped with boron concentration of 10^{16} cm^{-3} and assumed to be fully ionized. It is exposed to light such that electron-hole pairs are generated throughout the volume of the bar at the rate of 10^{20} cm^{-3} s^{-1}. If the recombination lifetime is 100 μs, intrinsic carrier concentration of silicon is 10^{10} cm^{-3} and assuming 100% ionization of boron, then the approximate product of steady state electron and hole concentrations due to this light exposure is

(A) $10^{20}cm^{-6}$ (B) $2 \times 10^{20}cm^{-6}$

(C) $10^{32}cm^{-6}$ (D) $2 \times 10^{32}cm^{-6}$

Q.2 The energy band diagram of a p type semiconductor bar of length L under equilibrium condition (i.e., the Fermi energy level E_F is constant) is shown in the figure. The valance band E_V is sloped since doping is non-uniform along the bar. The difference between the energy levels of the valence band at the two edges of the bar is Δ.



If the charge of an electron is q, then the magnitude of the electric field developed inside this semiconductor bar is

(A) $\frac{\Delta}{qL}$ (B) $\frac{2\Delta}{qL}$

(C) $\frac{\Delta}{2qL}$ (D) $\frac{3\Delta}{2qL}$

Q.3 A single crystal intrinsic semiconductor is at a temperature of 300 K with effective density of states for holes twice that of electrons. The thermal voltage is 26 mV. The intrinsic Fermi level is shifted from mid-bandgap energy level by

(A) 18.02 meV

(B) 9.01 meV

(C) 13.45 meV

(D) 26.90 meV

Q.4 Consider the recombination process via bulk traps in a forward biased pn homojunction diode. The maximum recombination rate is U_{max}. If the electron and hole capture cross-sections are equal, which one of the following is FALSE?

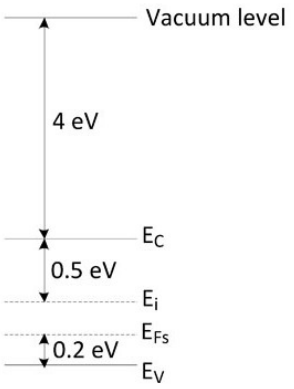
(A) With all other parameters unchanged, U_{max} decreases if the intrinsic carrier density is reduced.

(B) U_{max} occurs at the edges of the depletion region in the device.

(C) U_{max} depends exponentially on the applied bias.

(D) With all other parameters unchanged, U_{max} increases if the thermal velocity of the carriers increases.

Q.5 The band diagram of a p-type semiconductor with a band-gap of 1 eV is shown. Using this semiconductor, a MOS capacitor having V_{TH} of –16 V, C'_{ox} of 100 nF/cm^2 and a metal work function of 3.87 eV is fabricated. There is no charge within the oxide. If the voltage across the capacitor is V_{TH}, the magnitude of depletion charge per unit area (in C/cm^2) is



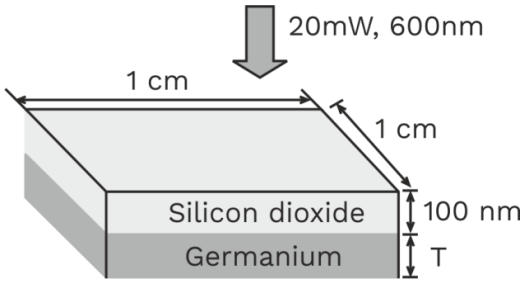
(A) 1.70×10^{-8}

(B) 0.52×10^{-8}

(C) 1.41×10^{-8}

(D) 0.93×10^{-8}

Q.6 A Germanium sample of dimensions 1cm × 1cm is illuminated with a 20 mW, 600 nm laser light source as shown in the figure. The illuminated sample surface has a 100 nm of loss-less Silicon dioxide layer that reflects one fourth of the incident light. From the remaining light, one-third of the power is reflected form the silicon dioxide-Germanium interface, one-third is absorbed in the Germanium layer, and one-third is transmitted through the other side of the sample. If the absorption coefficient of Germanium at 600 nm is $3\times10^4 cm^{-1}$ and the bandgap is 0.66 eV, the thickness of the Germanium layer, rounded off to 3 decimal places, is _____ µm.



Q.7 The intrinsic carrier concentration of silicon at 300 k is $2.5 \times 10^{15}/\mathrm{cm}^3$, After the p-type doping in the silicon, if the minority carrier density is $5 \times 10^{11}/\mathrm{cm}^3$, then the majority carrier density is-

(A)$6.25 \times 10^{19}/\mathrm{cm}^3$

(B)$2.5 \times 10^{20}/\mathrm{cm}^3$

(C)$1.25 \times 10^{19}/\mathrm{cm}^3$

(D)$0.8 \times 10^{19}/\mathrm{cm}^3$

Q.8 An n-type silicon bar is 0.5 cm long and $200\mu m^2$ in cross-sectional area. If the carrier mobility is $0.20\ \mathrm{m}^2/\mathrm{V}-\mathrm{s}$, the majority carrier concentration is $2.5 \times 10^{20}/\mathrm{m}^3$ and the charge of an electron is 1.6×10^{-19} coulomb, then the resistance of the silicon bar is _____ × 10^6 ohm. (Up to three decimal places)

Q.9 An n-type semiconductor has an intrinsic carrier concentration of $1.5 \times 10^{10}/\mathrm{cm}^3$ and an n-type doping concentration of $4.5 \times 10^{15}/\mathrm{cm}^3$. If the ratio of mobility of hole to that of electron ratio is 0.75, then the ratio $\frac{\sigma_i}{\sigma_n}$ is _____ $\times 10^{-6}$ (Up to two decimal places)

Q.10 Find the concentration of electrons in a p-type germanium semiconductor at 300 K. The conductivity is $100\ \Omega-\mathrm{cm}$, the mobility of holes is 1800 cm^3/V-s and the intrinsic carrier concentration is 3.5 x $10^{15}/cm^3$.

(A)$2.325 \times 10^{15}/\mathrm{cm}^3$

(B)$3.47 \times 10^{17}/\mathrm{cm}^3$

(C)$7.623 \times 10^{12}/\mathrm{cm}^3$

(D)$3.53 \times 10^{13}/\mathrm{cm}^3$

Q.11 A sample of Ge is doped to the extent of 10^{16} donor atoms / cm^3 and 5 × 10^{13} acceptor atoms /cm^3. At room temperature, the resistivity of pure Ge is 50 Ω-cm. If the applied electric field is 2 V/cm, mobility of holes is 1800 cm^2 /V-sec and mobility of electrons is 3800 cm^2/V-sec, then

(A)Concentration of holes $= 0.5 \times 10^{11}/\ \mathrm{cm}^3$

(B)Concentration of holes $= 9.96 \times 10^{15}/\mathrm{cm}^3$

(C) Total current density $= 15.21\ \mathrm{A}/\mathrm{cm}^3$

(D) Total current density $= 12.12\ \mathrm{A}/\mathrm{cm}^3$

Q.12 The Hall Effect is used to determine the mobility of holes in a p type silicon bar. Assume that the bar has resistivity $500\ \Omega-\mathrm{cm}$ and the magnetic field $(B_z)= 0.1\ Wb/m^2$. Hall current and Hall voltage are 10mA and 50mV respectively. If $d = w = 3\ \mathrm{mm}$, then mobility of holes(in cm^2/V-sec) is _____

Q.13 If the hole mobility of a semiconductor is

$$300\ \mathrm{cm^2/V-sec}\ and\ \frac{KT}{q}=26\mathrm{mV}$$

, then the hole diffusion constant $(in\ \ \mathrm{cm^2/sec})$ is _____ (Up to one decimal place)

Q.14 Consider the following statements regarding "Mass action law".

1. According to mass action law, ${n_0}{p_0} = {n_i}^2$ for a semiconductor at thermal equilibrium.

2. It is valid for degenerative semiconductors also.

3. It is valid at a constant temperature.

Which of the above statement(s) is/are true?

(A)1 only (B)1 and 2 only

(C)1 and 3 only (D)1, 2 and 3

Q.15 A semiconductor sample is uniformly doped with a donor impurity concentration of 5 × 10^{28} atoms / cm^3. The resistivity of the doped sample is $2\times10^{-8}\Omega-\mathrm{cm}$ at room temperature. If an electric field of $2\ \mathrm{V/cm}$ is applied across the sample, then the average drift velocity of electrons in the sample is -

(A)$7.5\times10^{-3}\ \mathrm{cm/sec}$

(B)$8.85\times10^{-3}\ \mathrm{cm/sec}$

(C)$12.5\times10^{-3}\ \mathrm{cm/sec}$

(D)$20\times5\times10^{-3}\ \mathrm{cm/sec}$

Q.16 An n-type silicon is having a doping carrier concentration of $N_D=10^{16}\ \mathrm{cm^{-3}}$. If this sample is uniformly illuminated, then the steady-state minority concentration in the sample will be $4\times10^4\ \mathrm{cm^{-3}}$. The rate at which the electron-hole pairs are generated is - (Assume $n_i=1.5\times10^{10}\ \mathrm{cm^{-3}}$ and minority carrier lifetime $\tau_p=10^{-6}\mathrm{sec}$)

(A)$1\times10^{10}\ \mathrm{cm^{-3}/s}$

(B)$1.25\times10^{10}\ \mathrm{cm^{-3}/s}$

(C)$1.75\times10^{10}\ \mathrm{cm^{-3}/s}$

(D)$2\times10^{10}\ \mathrm{cm^{-3}/s}$

Q.17 During an electron transition across the energy gap in an indirect bandgap semiconductor like silicon-

(A)The momentum of the electron changes.

(B)The direction of the motion of the electron changes.

(C)The potential energy (PE) of the electron changes.

(D)The kinetic energy (KE) of the electron changes.

Q.18 The intrinsic carrier concentration of silicon $(\mathrm{Eg}=1.12\mathrm{eV})$ at 300 K is $1.5\times10^{10}\ \mathrm{cm^{-3}}$. The value of n_i for Si at 400 K is-

(k = 8.62 × 10^{-5} eV/K)

(A)1.5 × 10^{10} cm^{-3}

(B)3.2 × 10^{11} cm^{-3}

(C)5.2 × 10^{12} cm^{-3}

(D)6.8×10^{13} cm^{-3}

Q.19 The intrinsic carrier concentration of silicon is $1.25 \times 10^{10}\ cm^{-3}$ and the doping concentration on the p-side and the n-side of the silicon diode are $2.5 \times 10^{15}\ cm^{-3}$ and $3 \times 10^{16}\ cm^{-3}$ respectively. If 0.2 volt of forward bias is applied to the diode, then the electron concentration at the edge of the depletion region on the p-side at $300\ K$ is _____ $\times 10^{8}\ cm^{-3}$. (Up to two decimal places)

[Assume V_T = 26 mV]

Q.20 In an n-type silicon, the donor concentration is 1 atom per 2.5×10^{6} silicon atoms. Assume that the effective mass of the electron is equal to the true mass. The temperature(in K) at which the Fermi level will coincide with the edge of the conduction band is _____. (Up to two decimal places)

Consider:

Concentration of silicon = 10×10^{20} atoms/cm^3

Boltzmann constant = 1.38×10^{-23} $m^2kgs^{-2}K^{-1}$

Plank's constant = 6.626×10^{-34} m^2kg/s

Effective mass of electron = 9.1×10^{-31} kg

Q.21 Find the probability of occupancy of a quantum state whose energy is 0.25 eV above the Fermi energy level.

(Assume T = 500 K, k = 1.38×10^{-23} J/K)

(A) 0.0030

(B) 0.006

(C) 0.0095

(D) 0.0459

Q.22 In a semiconductor diode, the cut-in voltage is the voltage at which-

(A)the depletion layer is formed.

(B)the current just starts to flow.

(C)the current is less than 1% of the maximum rated current.

(D)the current reaches its saturation point.

Q.23 A semiconductor bar has a length of 6 cm, and a 12 V DC voltage is applied across it. If the mobility of electrons is 1200 cm^2/V-sec, then the average collision time of electrons is equal to _____ msec. (Up to two decimal places)

Q.24 An n-type silicon bar has an electron concentration of 1.5×10^{16} cm^{-3} in the Hall experiment. The Hall coefficient is-

(A)268.8 cm^3/C

(B)572.2 cm^3/C

(C)416.6 cm^3/C

(D)245.5 cm^3/C

Q.25 A semiconductor sample is uniformly doped with donor impurity concentration of 6×10^{28} atoms/cm^3. The resistivity of the doped sample is $1.5 \times 10^{8}\ \Omega$-cm at room temperature. If an electric field of 3 V/cm is applied across the given sample, the average drift velocity of the

electrons in the sample is _____× 10^{-3} cm/sec.

(up to two decimal places)

Q.26 An n-type silicon has doping carrier concentration of 10^{14} cm^{-3}, the intrinsic carrier concentration of 1.5 × 10^{10} $cm^{-3,}$ and the minority carrier lifetime of 10^{-6} sec. If the sample is uniformly illuminated and the steady-state minority carrier concentration in the sample is 3 × 10^{6} cm^{3}, then

(A) Concentration of holes = 2.25 × 10^{6} cm^{-3}

(B) Concentration of holes = 1.5 × 10^{6} cm^{-3}

(C) Generation rate of Electron-hole pairs = 36 × 10^{10} cm^{-3}/sec

(D) Generation rate of Electron-hole pairs = 75 × 10^{10} cm^{-3}/sec

Q.27 A compensated n-type silicon at T = 300 K has acceptor and donor concentrations of 10^{17} cm^{-3} and 3 × 10^{17} cm^{-3} respectively. If the mobility of electrons is 400 cm^2/V-sec, then the conductivity of the silicon sample is _____ $(\Omega\text{-cm})^{-1}$ (Up to one decimal place)

Q.28 In a Hall experiment, a p-type semiconductor with hole concentration X_1 and Hall voltage V_{H1} is used. If the given p type sample is now replaced by another p type sample with hole concentration of X_2 (Where $2X_2 = 5X_1$), then the new Hall voltage V_{H2} is-

(A)0.4 V_{H1}

(B)2.5 V_{H1}

(C)5 V_{H1}

(D)1.6 V_{H1}

Q.29 A semiconductor is having electron mobility μ_n = 5000 cm^2/V-sec and the hole mobility μ_p = 300 cm^2/V-sec. If the hole concentration at minimum conductivity is 68 × 10^{11} cm^{-3}, then the intrinsic carrier concentration (in cm^{-3}) is-

(A)1.22 × 10^{11}

(B)12.22 × 10^{11}

(C)16.66 × 10^{11}

(D)1.66 × 10^{11}

Q.30 The resistance of a silicon semiconductor having area 75 μm^2, length 300 μm, intrinsic carrier concentration 1.5 × 10^{10} cm^{-3}, electron mobility 1350 cm^2/V sec, and hole mobility 500 cm^2/V-sec will be _____$M\Omega$. (Rounded off to the nearest integer)

Q.31 In a germanium sample at room temperature (T = 300 K), the donor concentration N_D = 5 × 10^{14} cm^{-3} and the acceptor concentration N_A = 10^{14} cm^{-3}. The difference between the fermi energy level and the intrinsic fermi level ($E_f - E_{fi}$) is _____× 10^{-3} eV. (Up to one decimal place)

[Assume kT = 0.026 eV, and n_i= 2.5 × 10^{13} cm^{-3}]

Q.32 The ratio of the current for a forward bias voltage of 0.068 V to the current for the same magnitude of reverse bias voltage for a germanium diode is _____. (up to to two decimal places)

[Assume V_T = 26 mV]

(A)-13.60

(B)26.54

(C)-11.3

(D)17.38

Q.33 Two semiconductor materials A and B have the same properties except bandgap energy. If the bandgap energy of material A is 1.0 eV and that of material B is 1.3 eV, then the ratio of the intrinsic concentration of material A to that of material B at a temperature T = 300 K is_____. (Rounded off to the nearest integer)

Q.34 In an n-type semiconductor, at room temperature, the mobility is 0.38 m^2/V-sec and the lifetime is 350 μs. The electron diffusion length will be _____mm. (Rounded off up to 2 decimal places)

[Assume V_T = 26 mV].

Q.35 Consider a semiconductor material with fermi energy of 6.5 eV. Find the temperature at which there is a 1 percent probability that a state of 0.5 eV energy below the fermi energy level will contain a hole.

Assume Boltzmann constant, k = 8.61 × 10^{-5} eV/K

(A)-1263.8 °K

(B)1263.8 °K

(C)1263.8 °C

(D)-1263.8 °C

Q.36 An n-type germanium semiconductor sample has a donor density of 1 × $10^{22}/m^3$. It is used in a Hall experiment. The magnetic field is 0.5 T and the current density is 800 A/m^2. Find the Hall voltage if the distance between surfaces is 5 mm.

(A)1.25 V

(B)1.25 mV

(C)1.25 µV

(D)None of the above

Q.37 A silicon semiconductor is having intrinsic concentration n_i = 1.5 × $10^{10}/cm^3$, N_D = 2 × $10^{11}/cm^3$ and N_A = 1 × 10^{11}/ cm^3 at a given temperature. Find the value of n_0 and p_0 (Assuming 100% ionization).

(A)n_0 = 1.022 × 10^{11} cm^{-3}, p_0 = 1.125 × 10^9 cm^{-3}

(B)n_0 = 1.022 × 10^{11} cm^{-3}, p_0 = 2.2 × 10^9 cm^{-3}

(C)n_0 = 1 × 10^{11} cm^{-3}, p_0 = 2.2 × 10^9 cm^{-3}

(D)n_0 = 1 × 10^{11} cm^{-3}, p_0 = 1.125 × 10^9 cm^{-3}

Q.38 A silicon semiconductor is doped with 1 × $10^{16}/cm^3$ boron atoms and the fermi level at 300K is 0.36 eV below E_{Fi}.

The donor concentration in the silicon semiconductor will be _____ × 10^{15} cm^{-3}. (Up to 2 decimal places)

Assume $n_i = 1 \times 10^9/cm^3$& 100% ionization

Q.39 For a semiconductor material $N_C = 1.5 \times 10^{18}/cm^3$ $N_V = 1.3 \times 10^{19}/ cm^3$ and $E_g = 1.43$ eV at T = 300 K. Which of the following statement(s) is/are correct?

(A)The fermi level is located 0.028 eV above the intrinsic fermi level.

(B)The fermi level is located 0.028 eV below the intrinsic fermi level.

(C)Intrinsic concentration of the semiconductor $n_i = 4.06 \times 10^6$ $/cm^3$

(D)Intrinsic concentration of the semiconductor $n_i = 4.06 \times 10^{12}/m^3$.

Q.40 An n-type semiconductor has $N_D = 1 \times 10^{13}/cm^3$ and carrier lifetime of electrons, $\tau_n = 5 \times 10^{-6}$ sec. If intrinsic carrier concentration $n_i = 1 \times 10^{10}/cm^3$ then, the carrier lifetime of holes is _____ psec. (Rounded off to the nearest integer)

Q.41 For an intrinsic semiconductor material, the center of the band gap is 0.4 eV at temperature T = 300 K. If the effective mass of a hole is four times that of the electron, then the fermi level will be-

(A)0.4267 eV

(B)0.3733 eV

(C)0.2267 Ev

(D)0.1733 eV

Q.42 The electron concentration in a silicon semiconductor decreases linearly from 1×10^{18} cm^{-3} to 8×10^{17} cm^{-3} over a distance of 0.2 cm. If the cross-section area of the semiconductor is 0.06 cm^2, then the electron diffusion current will be _____mA. (Rounded off to the nearest integer)

Assume electron diffusion constant, $D_n = 35$ cm^2/sec

Q.43 Which of the following statement(s) is/are **correct** regarding Fermi level of a semiconductor?

(A)If the Fermi level is below the intrinsic level, then the semiconductor is called p-type compensated semiconductor.

(B)In a highly degenerate n-type semiconductor, the Fermi level lies in the conduction band.

(C)The Fermi Dirac distribution function is given as

$$f(E) = \frac{1}{1 - e^{(E-E_F)/kT}}$$

Where, E_F is Fermi energy level.

(D)If the Fermi level is above the intrinsic level, then the semiconductor is called an n-type compensated semiconductor.

Q.44 Consider a Ge semiconductor at T = 300 K in which $N_A = 3 \times 10^{15}/cm^3$ and $N_D = 10^{16}/cm^3$. Assume $n_i = 1.5 \times 10^{10}/cm^3$.

Calculate hole concentration at thermal equilibrium for the Ge semiconductor.

(A)$324 \times 10^4/cm^3$

(B)$3.214 \times 10^4/cm^3$

(C)$32.14 \times 10^4/cm^3$

(D)$0.324 \times 10^4/cm^3$

Q.45 Evaluate the built-in voltage and depletion region width for the Si diode if the donor atom's and acceptor atom's concentration are $N_D = 4\times10^{16}$ cm^{-3} and $N_A = 10^{16}$ cm^{-3} respectively. Assume room temperature operation with $V_T = 0.025$ V, $n_i = 1.5 \times 10^{10}$ cm^{-3} and $\epsilon = 1.04 \times 10^{-12}$ F/m.

(A)$V_0 = 0.7$ V and W = 33.72 µm

(B)$V_0 = 0.5$ V and W = 3.372 µm

(C)$V_0 = 0.1135$ V and W = 0.282 µm

(D)$V_0 = 0.12$ V and W = 0.011375 m

Q.46 Calculate the fermi energy levels on the n-side and p-side with respect to intrinsic Fermi level for a silicon P-N junction at zero biasing and doping concentrations of $N_A = 10^{17}$ cm^{-3} and $N_D = 10^{15}$ cm^{-3} at T = 300K. Consider kT = 0.0259 eV and $n_i = 1.5 \times 10^{10}$ cm^{-3}.

(A)$E_{n\text{-side}} = 0.0228$ eV and $E_{p\text{-side}} = 0.0347$ eV

(B)$E_{n\text{-side}} = 0.287$ eV and $E_{p\text{-side}} = 0.407$ eV

(C)$E_{n\text{-side}} = 0.0028$ eV and $E_{p\text{-side}} = 0.131$ eV

(D)$E_{n\text{-side}} = 0.013$ eV and $E_{p\text{-side}} = 0.0287$ eV

Q.47 Consider a Si intrinsic semiconductor. Let the intrinsic carrier concentration be n_i. If the energy gap is constant and n_i is tripled, then the factor by which temperature has to be changed is _____. (Up to 2 decimal places)

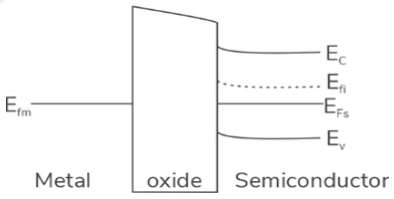
Q.48 Consider an n-type silicon semiconductor with dopants concentration $N_D = 2\times10^{15}/cm^3$. At a temperature 'To', if only 50% of dopants are ionized, then the concentration of holes is $p\times10^{15}/cm^3$. The value of p (integer) is-

[Assume $n_i = 2.5 \times 10^{15}/cm^3$].

(A)1 (B)2

(C)3 (D)4

Q.49 Consider the band diagram of a metal oxide semiconductor shown below.



The mode of operation in this MOS is-

(A)Accumulation

(B)Flat Band

(C)Inversion

(D)Depletion

Q.50 Consider an n-type semiconductor in thermal equilibrium. Donor concentration varies exponentially $n(x) = N_0 e^{-\frac{x}{d}}$ over the range $0 \leq x \leq d$, where N_0 is constant. Then which of the following is/are correct?

(A) $E(x) = \frac{V_T}{d}, 0 \le x \le d$

(B) $E(x) = -\frac{V_T}{d}, 0 \le x \le d$

(C) $V(x) = V_T, 0 \le x \le d$

(D) $V(x) = -V_T, 0 \le x \le d$

Q.51 A doped semiconductor has dopant concentration of $N_d = 2.5 \times 10^{14}$ and $N_a = 1 \times 10^{14}$ at T = 300K (Given $n_i = 2.5 \times 10^{14}$).

Select the correct option(s).

(A)The material is p-type.

(B)The material is n-type.

(C)The value of p_o is 1.86×10^{14} cm^{-3}.

(D)The value of n_o is 3.36×10^{14} cm^{-3}.

Q.52 A silicon sample is doped with 1 donor atom per 4×10^8 Si atoms. At what temperature will E_F coincide with E_C? [Given that Si has 10×10^{22} atoms/cm^3 and effective mass is the same as true mass.]

(A)0.14° K (B)3.14° K

(C)2° K (D)5.1°K

Q.53 Two semiconductor materials at room temperature have the same properties except that material-1 has a band gap of 1.2 eV and material-2 has a band gap of 1.36 eV. The ratio of intrinsic concentration of material-1 to material-2 is

[Assume kT = 26 meV]

(A)2.157 (B)4.655

(C)21.69 (D)49.65

Q.54 The maximum resistivity of a silicon sample is 7 kΩ-cm. The electron mobility is 1600 cm^2/V-sec and the mobility of holes is $400\ \text{cm}^2/\text{V} - \text{sec}$. The intrinsic carrier concentration of the Si sample is-

(A)2.32×10^{10} /cm^3

(B)4.75×10^{10}/ cm^3

(C)3.62×10^{11}/cm^3

(D)5.58×10^{11}/cm^3

Q.55 The donor concentration in an n type semiconductor held at T = 300 K has an x dependence given by

$N_d(x) = 10^{15} - 10^{18} \times \text{cm}^{-3}$

where, x is in centimeters and varies between $0 \le x \le 1$ µm. the resulting field Ex at x = 0 is _____ V/cm.

Q.56 Which of the metals listed below will you choose so as to make an ohmic contact to n-type InAs?

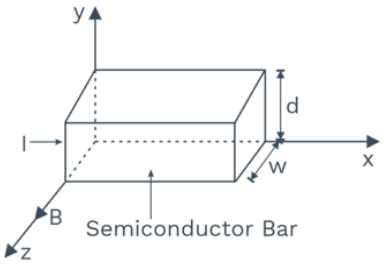
(A)Gold

(B)Silver

(C)Copper

(D)Aluminium

Q.57 Consider the given figure of an n-type semiconductor bar.



The parameters for the Hall experiment setup are:

Electric field = 500 V/m

B_z = 10.5 Wb/m^2

w = 10 mm

d = 5 × 10^{-2} m

Mobility: μ_n = 1.2 m^2/V-sec

The Hall voltage (V_H) is _____ volts.

Q.58 The band gap of silicon at 50°C is-

(A)1.2 eV

(B) 0.92 eV

(C)1.09 eV

(D)0.7 eV

Q.59 A germanium diode is biased with a current of 10 mA. The value of current change, if the diode voltage V_D changes by 2 mV is _____ mA. (V_T = 25 mV)

(Rounded off to one decimal place)

Q.60 A Si diode is operating at room temperature with forward voltage of 800 mV and has a saturation current of 30 nA . Its dynamic resistance is-

(A)26.67 MΩ

(B)0.36 Ω

(C)26.67 Ω

(D)36 Ω

Q.61 In a particular sample of n-type semiconductor, the donor concentration dependence on x at 300 K is given by

$N_d(x) = 10^{16} - 10^{20}x$

With x is in the range 0 ≤ x ≤ 1.5 µm. Calculate the resulting field EX at x = 0.5 µm is _____ v/cm.

Q.62 If a silicon sample is doped with 2 × 10^{16} phosphorous atom/cm^3 and 6 × 10^{16} boron atom/cm^3 with full ionization, the material is

(Consider n_i = 1.5 × 10^{10})

(A)n-type with carrier concentration of 4×10^{16} atom/cm^3

(B)p-type with carrier concentration of 4×10^{16} atom/cm^3

(C)n-type with carrier concentration of 2×10^{16} atom/cm^3

(D)p-type with carrier concentration of 2×10^{16} atom/cm^3

Q.63 An n-type semiconductor has following parameters:

Doping concentration = 3 × 10^{16} atom/cm^3

Intrinsic concentration = 2.5 × 10^{10} atom/cm^3

$\mu_n = 3500\ \text{cm}^2/\text{V} - \text{sec}$ and

$\mu_p = 1750\ \text{cm}^2/\text{V} - \text{sec}$

What is the ratio of conductance of n-type semiconductor to intrinsic conductance?

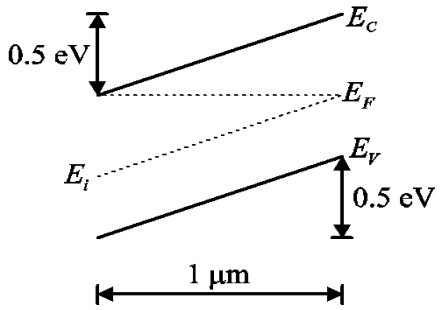
(A)0.6×10^{-5}

(B)8×10^5

(C)25×10^6

(D)7.5×10^6

Q.64 Consider the equilibrium energy band diagram shown in the figure below.



The value of the effective electric field for electrons (in kV/cm) is _____. (in integer)

Q.65 A silicon sample is doped with 1 donor atom per 4×10^8 Si atoms. At what temperature will E_F coincide with E_C? Given that Si has 10×10^{22} atom /cm^3 and effective mass is the same as true mass.

(A)0.14° K

(B)3.14° K

(C) 2° K

(D)5.1°K

Q.66 Find out the hole concentration in the semiconductor having the following parameters. Consider minimum conductivity.

$\mu_n = 1000 \text{ cm}^2/\text{V} - \text{s}$
$\mu_p = 600 \text{ cm}^2/\text{V} - \text{s}$
$\text{n}_\text{i} = 2.5 \times 10^{13} \text{ cm}^{-3}$

(A)$1.5 \times 10^{13} \text{ cm}^{-3}$

(B)$3.22 \times 10^{13} \text{ cm}^{-3}$

(C)$3.22 \times 10^{13} \text{ m}^{-3}$

(D)$2.5 \times 10^{16} \text{ cm}^{-3}$

Q.67 An intrinsic semiconductor is characterized by the relation of effective masses of electrons and holes as $m_n = 0.5m_p$ (at room temperature). Find out the location of intrinsic Fermi level with respect to the midpoint of the energy bandgap in the semiconductor. (Given kT= 0.026 eV)

(A)0.035 eV below the midpoint of the energy gap.

(B)0.013 eV above the midpoint of the energy gap.

(C)0.013 eV below the midpoint of the energy gap.

(D)–0.035 eV above the midpoint of the energy gap.

Q.68 Which of the following statement(s) is/are **incorrect**?

(A)A n-type semiconductor behaves as an extrinsic semiconductor at very high temperature.

(B)A n-type semiconductor behaves as an intrinsic semiconductor at very high temperature.

(C)A p-type semiconductor behaves as an extrinsic semiconductor at very high temperature.

(D)Both n & p-type semiconductors behave as intrinsic semiconductors at very high temperature

Q.69 For an extrinsic semiconductor, select the **false** statements.

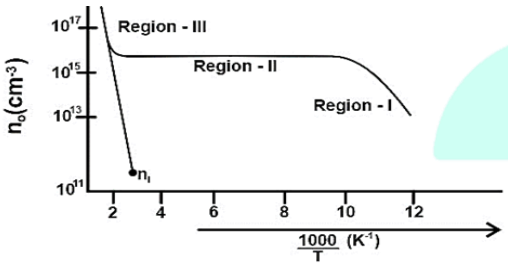
(A)In a highly degenerate n-type semiconductor, the Fermi level is in the conduction band.

(B)In an n-type semiconductor, the position of the fermi-level is above than the centre of the energy gap.

(C)In a p-type semiconductor, the position of the fermi-level is above the centre of the energy gap.

(D)In a p-type semiconductor, the position of the fermi-level can be anywhere.

Q.70 Figure shows the carrier concentration versus inverse temperature plot for Silicon with about 10^{15} donors/cm^3. The names of three regions mentioned as region I, region II and region III in fig are:



(A)Ionization – region, Extrinsic-region, Intrinsic-region

(B)Intrinsic-region, Extrinsic-region, Ionization – region

(C)Extrinsic-region, Ionization – region, Intrinsic-region

(D)Extrinsic-region, Intrinsic-region, Ionization – region

Q.71 A resistor measures 8Ω at 30° and 10Ω at 60°C. What value of the resistor will be measured at 0°C.

(A)3Ω (B)4Ω

(C)5 Ω (D)6 Ω

Q.72 The current density of n-type germanium is $50\ \mathrm{A/m^2}$ and the resistivity of $0.6\Omega - \mathrm{m}$. If the mobility of electron is 0.35 m^2-V-sec, the drift velocity is _____m/sec.

(A)20 (B)17.5

(C)10.5 (D)15.5

Q.73 Two semiconductor materials have same properties except that material-1 has band gap at 0°K as 1.2eV and material-2 has a band gap at 0°K as 1.21eV. The ratio of square of intrinsic concentration of material-2 to material 1 is _____. (Given KT = 0.025eV)

(A) 0.0145 (B)0.67

(C)1.492 (D)0.09

Q.74 The hall coefficient in an n-silicon bar is given by 3 x 10^3 cm^3/C. The electron concentration in the bar is given by-

(A)1.25 x 10^{15}

(B)2.08 x 10^{15}

(C)3.33 x 10^{-2}

(D)3.33 x 10^{15}

Q.75 Assume n_i= 1.5 x 10^{10}/cm^3 for silicon. In a p-type silicon sample, the accepter concentration at 300 K is 225 x 10^{14}/cm^3. The hole and electron concentrations in the sample will be-

(A)p = 225 x 10^{14}/cm^3 , n = 10^4/cm^3

(B)$p > 225 \times 10^{14}/cm^3$, $n < 10^4/cm^3$

(C)$p < 225 \times 10^{14}/cm^3$, $n < 10^4/cm^3$

(D)$p < 225 \times 10^{14}/cm^3$, $n > 10^4/cm^3$

Q.76 For a semiconductor, intrinsic carrier concentration at 300°K is given as $10^{10}/cm^3$. Find the intrinsic concentration at 600°K.

(A)$4 \times 10^{10}/cm^3$

(B)$2 \times 10^{10}/cm^3$

(C)$8 \times 10^{10}/cm^3$

(D)$2.83 \times 10^{10}/cm^3$

Q.77 The continuity equation is also known as the law of conservation of _____.

(A)charge (B)momentum

(C)power (D)energy

Q.78 The ratio of the diffusion coefficient to the mobility in a semiconductor has the unit-

(A)V (B)V/cm

(C)cm/V (D)V^{-1}

Q.79 Drift current in a semiconductor depends upon

(1) Carrier concentration gradient

(2) Electric field

(3) Mobility

(4) Carrier concentration

Which of the following are correct?

(A)1 and 2 only

(B)2 and 3 only

(C)1, 2 and 3 only

(D)2, 3 and 4 only

Q.80 Consider the following statements, if electric field is very high and increasing, then the

1. Velocity of charge carrier is constant.
2. Velocity of charge carrier increases.
3. Mobility of charge carriers increases.
4. Mobility of charge carrier decreases.

Which of the following are incorrect

(A)1 and 3 only

(B)1 and 4 only

(C)2 and 3 only

(D)2 and 4 only

Q.81 The cut-off wavelength of light that can be used for intrinsic excitation of a semiconductor material with bandgap, $\mathrm{E}_g = 1.2\mathrm{eV}$ is _____.

(A)$103\mu\mathrm{m}$

(B)$10.33\mu\mathrm{m}$

(C)$1.03\mu\mathrm{m}$

(D)$10.33\ \mathrm{A}^\circ$

Q.82 If in a certain magnetic material, the applied magnetic field is 10A/m, magnetization is 60 A/m and molecular field constant (γ) is 0.2, then the internal field is-

(A)10 A/m

(B)32 A/m

(C)3.2 A/m

(D)8.6 A/m

Q.83 For an n-type semiconductor, the impurity level is 50 meV below the conduction band. In a thermal collision,

the transferrable energy is kT. The value of T for which the electrons start to jump in the conduction band is -

(A)46.4 K (B)232 K

(C)580 K (D)464 K

Q.84 Consider the following statements:

1. Hall effect is used to determine whether the semiconductor is p-type or n-type.

2. Hall effect is used to determine carrier mobility.

3. Hall effect is used to calculate drift velocity

Which of the above statements are not correct?

(A)1 and 2

(B)2 and 3

(C)1, 2 and 3

(D)None of these

Answer Key

Q.1 Ans. [D]
Q.2 Ans. [A]
Q.3 Ans. [B]
Q.4 Ans. [B]
Q.5 Ans. [A]
Q.6 Ans. [0.231, Range 0.230 to 0.232]
Q.7 Ans. [C]
Q.8 Ans. [3.125, Range 3.1 to 3.2]
Q.9 Ans. [5.83, Range 5.8 to 5.86]
Q.10 Ans. [D]
Q.11 Ans. [A, D]
Q.12 Ans. [300, Range 299 to 301]
Q.13 Ans. [7.8, Range 7.75 to 7.85]
Q.14 Ans. [C]
Q.15 Ans. [C]
Q.16 Ans. [C]
Q.17 Ans. [A, C, D]
Q.18 Ans. [C]
Q.19 Ans. [1.37, Range 1.35 to 1.39]
Q.20 Ans. [0.19, Range 0.18 to 0.2]
Q.21 Ans. [A]
Q.22 Ans. [C]
Q.23 Ans. [2.5, Range 2.45to 2.55]
Q.24 Ans. [C]
Q.25 Ans. [20.82, Range 20.5 to 21]
Q.26 Ans. [A, D]
Q.27 Ans. [12.8, Range 12.7 to 12.9]
Q.28 Ans. [A]
Q.29 Ans. [C]
Q.30 Ans. [9009, Range 9008 to 9010]
Q.31 Ans. [72.2, Range 72 to 72.4]
Q.32 Ans. [A]
Q.33 Ans. [327, Range 325 to 329]
Q.34 Ans. [1.86, Range 1.81 to 1.9]
Q.35 Ans. [B]
Q.36 Ans. [B]
Q.37 Ans. [B]
Q.38 Ans. [8.89, range 8.85 to 8.93]
Q.39 Ans. [A, C, D]
Q.40 Ans. [5]
Q.41 Ans. [A]
Q.42 Ans. [-336, Range -340 to -332]
Q.43 Ans. [A, B, D]
Q.44 Ans. [B]
Q.45 Ans. [A]
Q.46 Ans. [B]
Q.47 Ans. [2.08, Range 2 to 2.14]
Q.48 Ans. [B]
Q.49 Ans. [A]
Q.50 Ans. [A, D]
Q.51 Ans. [B, C, D]
Q.52 Ans. [A]
Q.53 Ans. [C]
Q.54 Ans. [D]
Q.55 Ans. [25.9, Range 25.5 to 26.4]
Q.56 Ans. [A, B, C, D]
Q.57 Ans. [315, range 313 to 317]
Q.58 Ans. [C]
Q.59 Ans. [0.8, range 0.75 to 0.85]
Q.60 Ans. [B]
Q.61 Ans. [518, range 516 to 520]
Q.62 Ans. [B]
Q.63 Ans. [B]
Q.64 Ans. [5]
Q.65 Ans. [A]

Q.66 Ans. [B]

Q.67 Ans. [B]

Q.68 Ans. [A, C]

Q.69 Ans. [C, D]

Q.70 Ans. [A]

Q.71 Ans. [D]

Q.72 Ans. [C]

Q.73 Ans. [B]

Q.74 Ans. [B]

Q.75 Ans. [B]

Q.76 Ans. [D]

Q.77 Ans. [A]

Q.78 Ans. [A]

Q.79 Ans. [D]

Q.80 Ans. [C]

Q.81 Ans. [C]

Q.82 Ans. [B]

Q.83 Ans. [C]

Q.84 Ans. [D]

2 P-N Junction Diode

MCQ, MSQ and NAT

Q.1 A PN junction diode has two types of capacitances: Junction capacitance and Diffusion capacitance. The junction capacitance depends on -

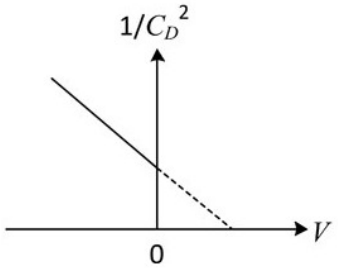
(A)Applied voltage

(B)Doping concentration

(C)Majority carrier lifetime

(D)Quiescent current of the diode

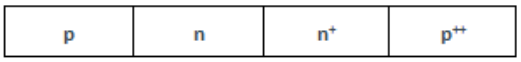
Q.2 A one-sided abrupt pn junction diode has a depletion capacitance C_D of 50 pF at reverse bias of 0.2 V. The plot of $1/C_D^2$ versus the applied voltage V for this diode is a straight line as shown in the figure below. The slope of the plot is _____ $\times 10^{20}$ $F^{-2}V^{-1}$.



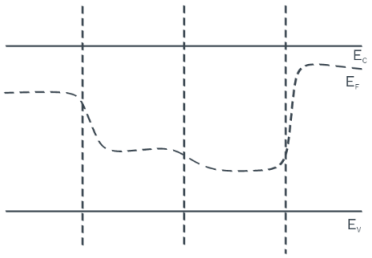
(A) –5.7 (B) –3.8

(C) –1.2 (D) –0.4

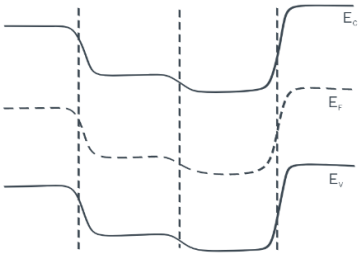
Q.3 Which one of the following options describes correctly the equilibrium band diagram at T= 300 K of a Silicon pnn^+p^{++} configuration shown in the figure?



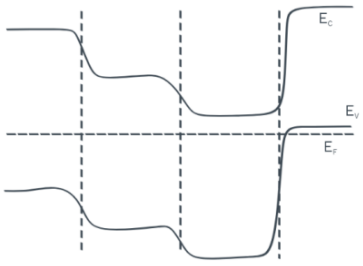
(A)



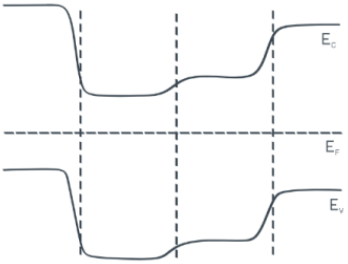
(B)



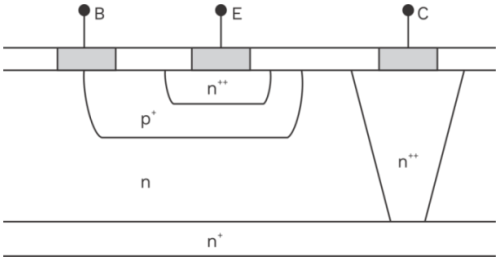
(C)



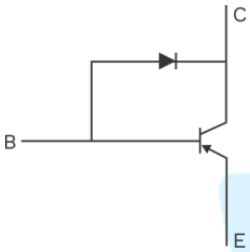
(D)



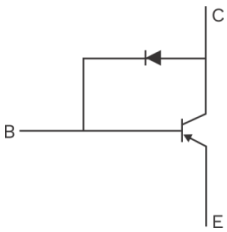
Q.4 The correct circuit representation of the structure shown in the figure is:



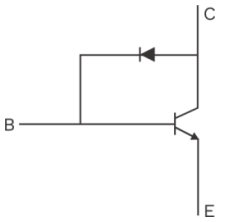
(A)



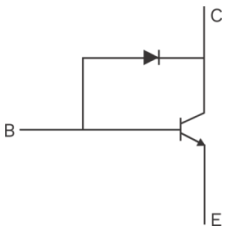
(B)



(C)



(D)



Q.5 In an ideal pn junction with an ideality factor of 1 at T=300 K, the magnitude of the reverse bias voltage required to reach 75% of its reverse saturation current, rounded off to 2 decimal places, is _____mV.

[k = 1.38×10^{-23} JK^{-1}, h = 6.625×10^{-34} J – s, q = 1.602×10^{-19}C]

Q.6 When the temperature increases, the dynamic resistance of a diode-

(A)increases

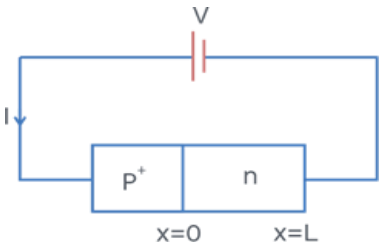
(B)decreases

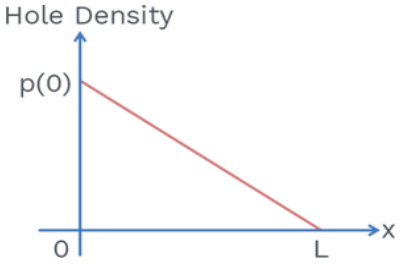
(C)does not change

(D)none of the above

Q.7 For a Germanium PN junction diode, the barrier voltage is 0.3 V. N_A = 4 × 10^{20} cm^{-3} and $N_D \gg N_A$. Given, A = 0.01 cm^2 and $\varepsilon = 141.584 \times 10^{-12}$ F/cm. The value of transition capacitance at reverse a bias voltage of 12 V is _____nF. (Up to 3 decimal places)

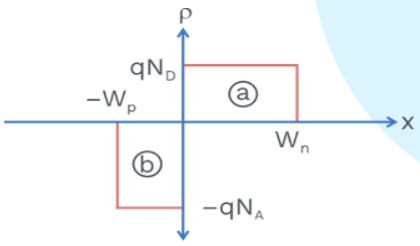
Q.8 For the diode shown in the figure below, the current in the diode is due to the diffusion of holes from x=0 to x=L. The injected hole concentration is linear as shown below.





The diffusion coefficient is 15 cm^2/sec, p(0) = $10^{12}/cm^3$ and L = 10^{-3} cm. The current density is_____mA/cm^2. (Up to one decimal place)

Q.9 Consider the following charge density distribution of the step graded junction of a PN junction diode under thermal equilibrium conditions.



If $W_p = 0.9\mu m$ and $W_n = 1.7\ \mu m$, then the ratio of the concentration of acceptor atoms to the concentration of donor atoms is_____. (Up to two decimal places)

Q.10 Two diodes D_1(Ge) and D_2(Si) have the same I_0. The effect of temperature on I_0 for both the diodes is also the same. In a circuit, D_1 is placed between two points [point (1) and point (2)]. D_1 carries a current of 8mA at a voltage of 0.6 V. Assume that the temperature is 27°C.

Now, D_1 is replaced by D_2. The current through diode D_2 (in nA) at 87°C is_____. (Up to 3 decimal places)

{Consider thermal voltage at 27°C to be V_T = 26 mV.}

Q.11 A step-graded silicon PN junction has a doping concentration given as $N_A = N_D = 10^{17}/\text{cm}^3$. The magnitude of the reverse bias voltage is 6.3 V and the peak electric field across the junction is $10^6\ \text{V/cm}$. Now, the magnitude of the reverse bias voltage is increased to 9.2V. The peak electric field across the junction reaches ____ $\times 10^6\ \text{V/cm}$. (Up to 3 decimal places)

Q.12 A silicon diode has diffusion capacitance of $0.5\mu\text{F}$ at room temperature. The forward current of the diode is 1 mA. If $N_A \gg N_D$, then the diffusion length of holes is _____cm. (Up to 3 decimal places)

(Consider hole mobility as 500 cm^2/V-sec).

Q.13 A Si diode operates at room temperature(27°C) with a forward voltage of 800 mV and has a saturation current of 30 nA. Its dynamic resistance is-

(A)$26.67\ \text{M}\Omega$

(B)$0.33\ \Omega$

(C)$26.67\ \Omega$

(D)$36\ \Omega$

Q.14 A diode is reverse biased with 8 V

and has junction capacitance of 15 pF. When the reverse bias is increased to 12 V, the capacitance drops to 13.05 pF. Consider V_0 as 0.6 V. The junction is -

(A)Linearly graded junction

(B)Step graded junction

(C)Both (A) and (B)

(D)None of the above

Q.15 A silicon diode operates at a fixed forward bias voltage of 0.7 V. If the temperature increases from 30°C to 80°C, then the factor by which the current gets multiplied is _____. (Up to one decimal place)

Q.16 In a silicon-based n^+p step junction diode, the reverse saturation current density is-

(A)directly proportional to the acceptor concentration

(B)directly proportional to the donor concentration

(C)inversely proportional to the donor concentration

(D)inversely proportional to the acceptor concentration

Q.17 The transition capacitance of an abrupt p-n junction diode is 30 pF at 9 V. The value by which the capacitance decreases for 1 V increase in the bias voltage is –

(A)3.07 pF

(B)5.31 pF

(C)1.54 pF

(D)1.53 F

Q.18 Consider a germanium p-n junction diode with applied voltage of 0.25 Volts, I_0 = 2 μA and T = 300 K. (Assume V_T = 0.026 Volts)

The dynamic forward resistance is-

(A)8.67 Ω (B)0.867 kΩ

(C)1.153 Ω (D)0.867Ω

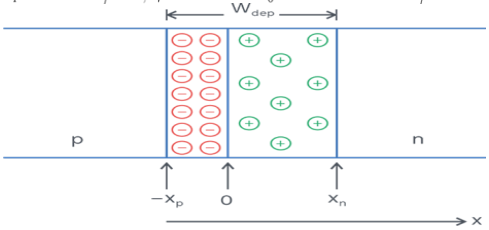
Q.19 A p-n junction is doped with $N_A = 3 \times 10^{18}$ cm^{-3} and $N_D = 1.5 \times 10^{17}$ cm^{-3} at 300 K. The built-in voltage, if $n_i = 1.5 \times 10^{10}$ cm^{-3}, is_____ V. (up to three decimal places)

[Assume V_T = 26 mV]

Q.20 Consider a p-n junction whose acceptor and donor concentrations are $10^{16}/cm^3$ and $10^{15}/cm^3$ respectively. The built-in voltage is 0.653 Volts.

Assume that

$n_i = 1 \times 10^{10}/cm^3, \epsilon_r = 11.7$ *and* $\epsilon_0 = 8.854 \times 10^{-14}$ F/cm



Which of the following statement(s) is/are correct regarding the given junction?

(A)The depletion layer width (W_{dep}) is 2.836 μm, when the junction is reverse biased with 5V.

(B)At reverse bias voltage, V_R = 5V, the magnitude of the charge stored on either

side of the junction is 1.65×10^{-13} C, if the junction area is 400 μm^2.

(C)The junction capacitance is 1.461×10^{-14} F, if the junction area is 400 μm^2.

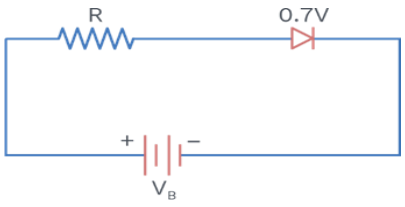
(D)The depletion layer width (W_{dep}) is 3.268 μm, when the junction is reverse biased with 5V.

Q.21 In a p-n junction diode, if the forward current is 0.063 A at a forward voltage of 0.6 V at (T = 300 K), then the reverse saturation current (in pA) is _____ (Up to two decimal places)

[Assume Boltzmann constant = 1.38×10^{-23} J/K and T is the absolute temperature]

Q.22 Consider a p-n junction diode in the given circuit that requires a minimum current of 2mA to be above the knee point of its V-I characteristics curve. The voltage across the diode is independent of the curve above the knee point.

The maximum value of R, so that the voltage is above the knee point when V_B = 5 Volts, is-



(A)2.15 kΩ (B)4.3 Ω

(C)4.3 kΩ (D)2.15 Ω

Q.23 The depletion width of an abrupt p-n junction is 0.532 μm and the penetration of the depletion region into the p-side is 0.00469 μm. The ratio of donor concentration on the n-side to the acceptor concentration on the p-side is _____ $\times 10^{-3}$. (Up to 3 decimal places)

Q.24 A one-sided abrupt junction has $10^{16}/cm^3$ of dopants on the lightly doped side. If the built-in potential is 0.2 V and zero bias voltage, then the depletion width of the abrupt junction is _____ μm. (Rounded off up to 3 decimal places)

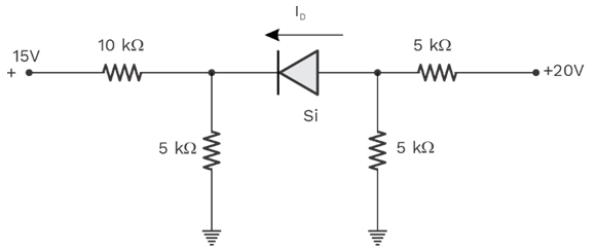
Assume q = 1.6×10^{-19} C

$\varepsilon_{Si} = 12$

$\varepsilon_0 = 8.85 \times 10^{-12}$ F/m

Q.25 In a PN junction, the depletion layer width is 3 μm at a reverse bias voltage of 1.2 V. At a reverse bias voltage of 7 V, the depletion layer width becomes double. The built-in potential of the junction is _____V. (Up to 2 decimal places)

Q.26 A diode circuit is shown in the figure below.



Assuming the diode cut-in voltage = 0.7 V, find the current (I_0) passing through the diode.

(A)0.737 mA (B)0.737 μA

(C)0.737 A (D)None of the above

Q.27 A uniformly doped pn junction at T = 300 K has only 20% of the total space charge region in the p region. If the built in potential V_{bi} = 0.9 V, then the acceptor impurity concentration N_A is _____ × 10^{13} cm^{-3}. (Up to 2 decimal places)

Assume n_i = 1.7 × 10^6 cm^{-3} and V_T =25.9 mV.

Q.28 Select the correct expression(s) for the width of the N-side in the P-N diode, if W is the width of the P-N junction, N_D is the donor concentration and N_A is the acceptor concentration.

(A) $W_N = \frac{N_A}{W\,(N_A + N_D)}$

(B) $W_N = \frac{N_D}{W\,(N_A + N_D)}$

(C) $W_N = \frac{WN_A}{N_A + N_D}$

(D) $W_N = \frac{W}{1 + \frac{N_D}{N_A}}$

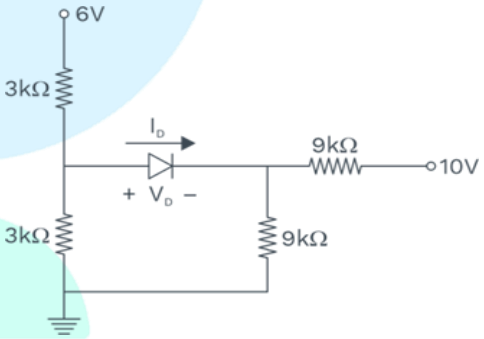
Q.29 A step-graded silicon diode has a donor concentration of N_D = 600 N_A and has acceptor concentration of N_A in the ratio of 1 : 10^6 atoms. The barrier voltage (in mV), assuming intrinsic carrier concentration n_i = 10^{10}/cm^3 and the total number of atoms as 5 × 10^{21} per cm^3, is _____(rounded off to the nearest integer)

[Consider V_T = 26 mV].

Q.30 A P-N junction diode at T = 300 has impurity doping concentration of N_A = 10^{16}/cm^3 and N_D = 5×10^{16}/cm^3. The ratio of transition capacitance at two different values of the reverse bias voltage is given as $\frac{C_T\,(at\ V_{r1})}{C_T\,(at\ V_{r2})} = 3$. If reverse bias voltage V_{r2} =17V, then the value of V_{r1} _____ V. [Rounded off to the nearest integer]

(Assume n_i = 10^8/cm^3 and V_T = 26mV).

Q.31 For the circuit shown below, which of the following statement(s) is/are correct?



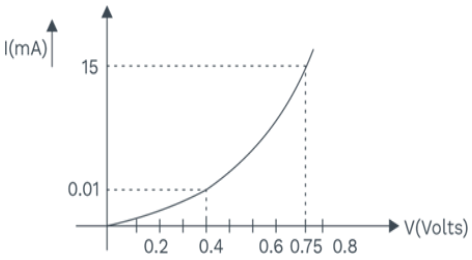
(A)The current through the diode is 0.33 A.

(B)Diode is always in ON state.

(C)Diode is OFF.

(D)Voltage across diode i.e., V_D is –2V.

Q.32 The V-I characteristics of a diode at room temperature is shown below. Find the value of η.

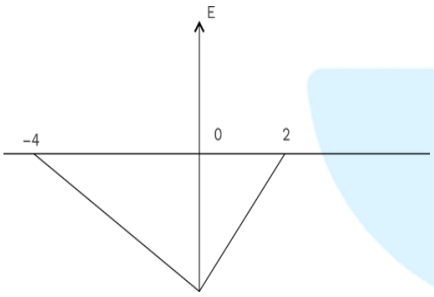


[Assume V_T = 26 mV.]

(A)1.32 (B)1.84

(C)2 (D)2.22

Q.33 The electric field distribution of a silicon abrupt PN junction in thermal equilibrium is-



Consider that the permittivity of silicon is 1.04×10^{-12} F/cm. The acceptor concentration of the junction is-

(A)4.875×10^{11} / cm^3

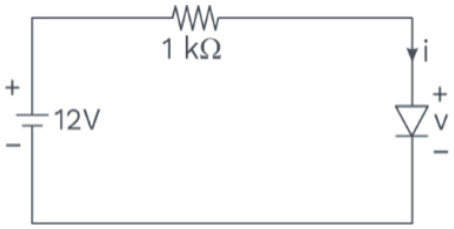
(B)4.875×10^{24} / cm^3

(C)2.43×10^{11} / cm^3

(D)2.43×10^{24} / cm^3

Q.34 If a p-n junction diode with carrier lifetime of holes (τ_p) equal to 15msec and mobility of holes equal to 3400 cm^2/V-sec is operated at room temperature then the ratio of diffusion coefficient (D_p) of the holes to the diffusion length (L_p) of holes (in cm/sec) is _____.(up to 1 decimal place) (Assume V_T = 0.025eV)

Q.35 A circuit containing diode is given below.



The relation between current and voltage is given by

$$i = \begin{cases} \frac{v-0.6}{480} A, & v \geq 0.6 \text{ V} \\ 0, & v < 0.6 \text{ V} \end{cases}$$

The current flowing in the circuit is _____ mA. (Up to two decimal places)

Q.36 In an abrupt p-n junction, the doping concentrations on the p and n side are

$N_A = 8 \times 10^{17}$/ cm^3 and $N_D = 1 \times 10^{18}$/ cm^3 respectively. The p-n junction is reverse biased, and the total depletion width is 5 μm. The depletion width (in μm) on the n-side is_____. (Up to 2 decimal places)

Q.37 Calculate the ratio of saturation current in germanium diode to that in silicon diode, if the germanium diode carries a current of 1mA at forward bias of 0.15V and silicon diode carries a current of 2mA at forward bias of 0.15V.

(Assume V_T = 25mV)

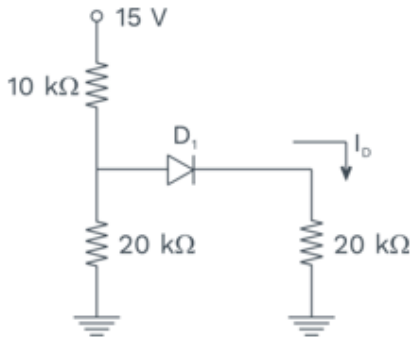
(A)42.18

(B)23.7×10^{-3}

(C)3.4×10^{-3}

(D)4.43×10^{3}

Q.38 If the forward biased voltage V_D = 0.7 V, then the diode current I_D is



(A)0.465 mA (B)0.349 mA

(C)0.476 mA (D)0.486 mA

Q.39 A silicon diode is biased with a current of 7 mA. The value of current change, if the diode voltage, V_D changes by 2mV is_____ mA. (V_T = 25mV)

(Rounded off to 2 decimal places)

Q.40 Consider a silicon p-n junction having concentration $N_A = 10^{17}$ cm^{-3} and $N_D = 10^{16}$ cm^{-3}. Assuming $n_i = 10^{11}$ cm^{-3}, the peak electric field for this junction is _____ (in kV/cm).

(Given kT = 0.026 eV and ε_{rSi} = 10).

(Answer up to 1 decimal place).

Q.41 Consider a silicon p-n junction at room temperature with doping concentration of $N_A = 3 \times 10^{16}$ cm^{-3} and $N_D = 9 \times 10^{15}$ cm^{-3}. (Assume that $n_i = 1.5 \times 10^{10}$ cm^{-3}, reverse bias voltage = 2V, cross section area = 10^{-4} cm^2)

Which of the following statement(s) is/are correct?

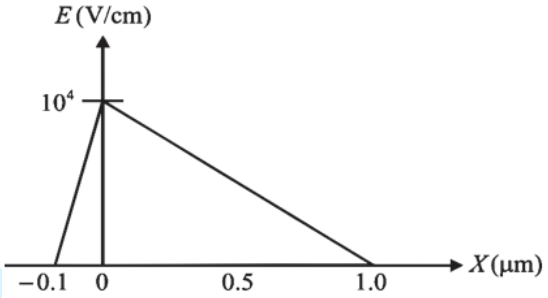
(A)Depletion width = 3.22×10^{-5}m

(B)Depletion width = 659 nm

(C)Transition capacitance = 1.34 pF

(D)Junction capacitance = 134 pF

Q.42 The electric field profile in the depletion region of a p-n junction in equilibrium is shown in the figure. Which one of the following statements is TRUE?



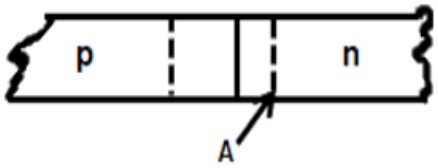
(A)The left side of the junction is n- type and the right side is p- type

(B)Both the n- type and p- type depletion regions are uniformly doped

(C)The potential difference across the depletion region is 550 mV

(D)If the p- type region has a doping concentration of 10^{15} cm^{-3}, then the doping concentration in the n- type region will be 10^{16} cm^{-3} .

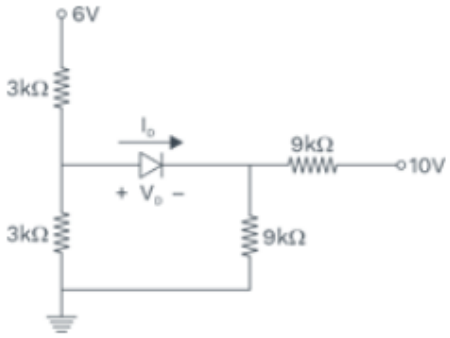
Q.43 A typical Si p-n junction is shown in Fig. At 300K the doping concentrations on the p and n sides are $N_a = 10^{15}$ cm^{-3} and $N_d = 10^{17}$ cm^{-3}, respectively. The mobility of electrons in the p-region is 1400 cm^2/V sec and the carrier life time = 10 μsec.



Assuming a case of low level injection, Find the total concentration of holes at A, when the supply of excess carriers of A is continuously replenished with time, under the application of a forward bias voltage of 0.5 V is _____ 10^{11} cm^{-3}. [Given: n_i = 1.5 x 10^{10} cm^{-3}]

Q.44 The built-in potential of a linear p n junction is 0.8V. If its junction capacitance at zero bias is 7pF, then the value of junction capacitance at 2V is _____ pF.

Q.45 For the circuit shown below. Which of the following statement(s) is/are correct?



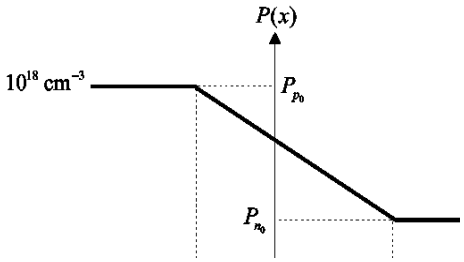
(A)The current through the diode is 0.33 A.

(B)Diode is always in the ON state.

(C)Diode is OFF.

(D)Voltage across the diode i.e. V_D = –2V.

Q.46 In a silicon p-n junction, the variation of the hole concentration in the space charge region is shown in figure below, where the hole concentration changes from P_{p_0} to P_{n_0} over the space charge layer width.



The hole diffusion constant in space charge region is $5\ \mathrm{cm}^2/\mathrm{sec}$. Then the hole diffusion current density J_p is

(A)$2.66\mathrm{kA/cm}^2$

(B)$3.66\mathrm{kA/cm}^2$

(C)$3.66\ \mathrm{mA/cm}^2$

(D)$2.66\ \mathrm{mA/cm}^2$

Q.47 A p-n junction has doping densities. N_a = 5 x 10^{18} cm^{-3} and N_d = 5 x 10^{15} cm^{-3} in the two regions. Assuming n_i = 1.5 x 10^{10} cm^{-3}, the built-in potential at 300 K is _____ volt.

Q.48 A PN junction diode is doped with $10^{16}cm^{-3}$, 10^{18} cm^{-3} in p and n side respectively. What is the ratio of penetration of depletion region in p side to that inside under open circuit condition

(A)10 (B)25

(C)100 (D)50

Q.49 A one-sided (p^+-n) step graded junction has 10^{14} per cm^3 of dopants on

the lightly doped side, zero bias voltage, and a built-in potential of 0.5V. The depletion width of the step graded junction is-

(given, $\in_s = 4$)

(A)1.5 µm (B)3 µm

(C) 2.8 µm (D)4 µm

Q.50 A potential barrier of 1 V exist across a pn junction, if the depletion width is $2X10^{-8}$m. What is the intensity of the electric field in this region.

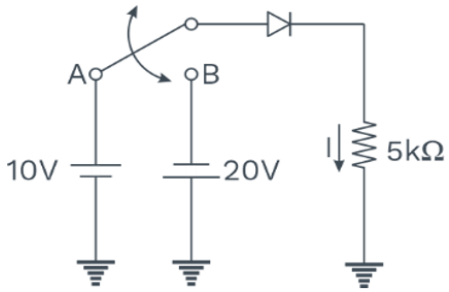
(A)5 x 10^7 V/cm

(B)2 x 10^8 V/cm

(C) 2 x 10^6 V/cm

(D) 5 x 10^5 V/cm

Q.51 Consider the circuit given below, the switch S is in position A initially and after reaching steady state switch is suddenly thrown into position B. The current I through the 5 kΩ resistor. Just after switch is connected to point B is



(A)0 Ma (B)–2 mA

(C)–4 mA (D)2 mA

Q.52 The built-in potential of an abrupt p-n junction is 0.8V. If its junction capacitance (C_j) at a reverse bias (V_R) of $1\ V$ is $6pF$. The value of junction capacitance at reverse bias of 4.6V is _____.

(A)3 pF (B)10.39 pF

(C)2.8 pF (D)3.46 pF

Q.53 In an abrupt p-n junction the doping concentrations on the p and n side are $N_A = 8 \times 10^{17}/cm^3$ and $N_D = 1 \times 10^{18}/cm^3$ respectively. The p-n junction is reverse biased and the total depletion width is 4µm. The depletion width on the p-side is_____?

(A)$1.78\mu m$ (B)$2\mu m$

(C)$2.22\mu m$ (D)$1.9\mu m$

Q.54 A junction is made between P—Si With doping density $N_{A1} = \frac{1}{e^2} \times 10^{15}\ cm^{-3}$ and P-Si with doping density $N_{A2} = e^2 \times 10^{15}\ cm^{-3}$. Assume 100% acceptor ionization. At room temperature, the magnitude of the built-in potential across this junction will be (Given V_T = 0.026V)

(A)6.5 mV (B)208 mV

(C)104 mV (D)52 mV

Q.55 Consider a silicon p-n junction at room temperature having the following parameters.

Doping on n-side $= 3 \times 10^{17}\ cm^{-3}$

Depletion width on p-side = 0.3 µm

Depletion width on n-side = 0.9 µm

Intrinsic carrier concentration $= 1.5 \times 10^{10}\ \mathrm{cm}^{-3}$

Dielectric constant of silicon = 12

The peak electric field in the device is

_____

(A) $4\mathrm{MVm}^{-1}$ (B) $8\mathrm{MVm}^{-1}$

(C) $8\mathrm{MVcm}^{-1}$ (D) $4\mathrm{MVcm}^{-1}$

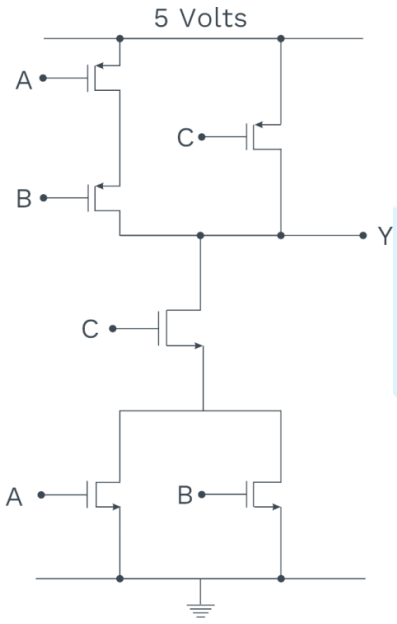
Answer Key

Q.1 Ans. [A ,B]
Q.2 Ans. MTA (Marks to all)
Q.3 Ans. [C]
Q.4 Ans. [D]
Q.5 Ans. [35.87, Range 34 to 38]
Q.6 Ans. [A]
Q.7 Ans. [191.924, Range 190 to 193]
Q.8 Ans. [2.4, Range 2.35 to 2.45]
Q.9 Ans. [1.88, Range 1.85 to 1.91]
Q.10 Ans. [0.512, Range 0.5 to 0.53]
Q.11 Ans. [1.186, Range 1.1 to 1.2]
Q.12 Ans. [0.018, Range 0 to 0.1]
Q.13 Ans. [B]
Q.14 Ans. [A]
Q.15 Ans. [4.8, Range 4.7 to 4.9]
Q.16 Ans. [D]
Q.17 Ans. [C]
Q.18 Ans. [D]
Q.19 Ans. [0.916, Range 0.88 to 0.93]
Q.20 Ans. [A, B, C]
Q.21 Ans. [5.35, Range 5.3 to 5.4]
Q.22 Ans. [A]
Q.23 Ans. [8.894, Range 8.8 to 8.9]
Q.24 Ans. [0.163, Range 0.16 to 0.17]
Q.25 Ans. [0.73, Range 0.71 to 0.75]
Q.26 Ans. [A]
Q.27 Ans. [11.95, Range 11.9 to 12]
Q.28 Ans. [C, D]
Q.29 Ans. [848, Range 845to 851]
Q.30 Ans. [1, Range 0.95 to 1.05]
Q.31 Ans. [C, D]
Q.32 Ans. [B]
Q.33 Ans. [A]
Q.34 Ans. [2380.5, Range 2375 to 2385]
Q.35 Ans. [7.7, Range 7.65 to 7.75]
Q.36 Ans. [2.22, Range 2.1 to 2.3]
Q.37 Ans. [B]
Q.38 Ans. [B]
Q.39 Ans. [0.28, Range 0.26 to 0.3]
Q.40 Ans. [46.5, Range 46 to 47]
Q.41 Ans. [B,C]
Q.42 Ans. [A, B, C, D]
Q.43 Ans. [5.45, Range 5.4 to 5.5]
Q.44 Ans. [4.6, Range 4.3 to 4.9]
Q.45 Ans. [C, D]
Q.46 Ans. [A]
Q.47 Ans. [0.838, Range 0.8 to 0.86]
Q.48 Ans. [C]
Q.49 Ans. [A]
Q.50 Ans. [D]
Q.51 Ans. [C]
Q.52 Ans. [D]
Q.53 Ans. [A]
Q.54 Ans. [C]
Q.55 Ans. [D]

3 Basics of BJT

MCQ, MSQ and NAT

Q.1 In the circuit shown



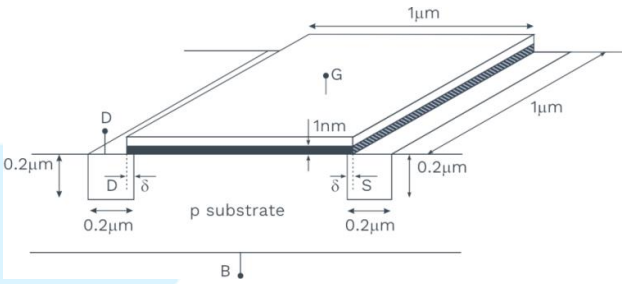
(A) $Y = \bar{A}\bar{B} + \bar{C}$

(B) $Y = (A + B)C$

(C) $Y = (\bar{A} + \bar{B})\bar{C}$

(D) $Y = (AB) + C$

Q.2 In the three-dimensional view of a silicon n-channel MOS transistor shown below, $\delta = 20$ nm . The transistor is of width $1\mu m$ .The depletion width formed at every p-n junction is 10 nm. The relative permittivity of Si and SiO_2, respectively, are 11.7 and 3.9, and $\varepsilon_0 = 8.9 \times 10^{-12}\ F/m$.

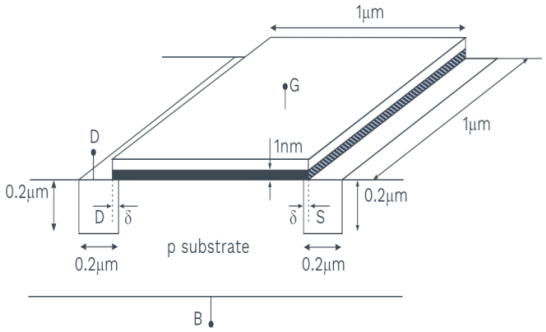


The gate- source overlap capacitance is approximately.

(A) 0.7 fF (B) 0.7 pF

(C) 0.35 fF (D) 0.24 pF

Q.3 In the three-dimensional view of a silicon n-channel MOS transistor shown below, $\delta = 20$ nm . The transistor is of width $1\mu m$ .The depletion width formed at every p-n junction is 10 nm. The relative permittivity of Si and SiO_2, respectively, are 11.7 and 3.9, and $\varepsilon_0 = 8.9 \times 10^{-12}\ F/m$.



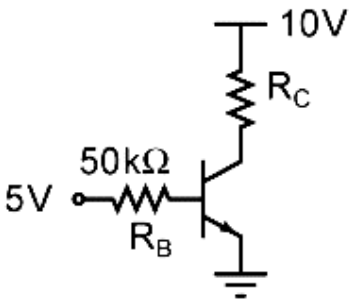
The source-body junction capacitance is Approximately

(A) 2 fF (B) 7 fF

(C) 2 pF (D) 7 pF

Q.4 A BJT is biased in forward active mode. Assume V_{BE} = 0.7 V, kT/q = 25 mV and reverse saturation current $I_S = 10^{-13}$ A. The transconductance of the BJT (in mA/V) is _____.

Q.5 In the circuit shown, the silicon BJT has β = 50. Assume V_{BE} = 0.7 V and $V_{CE(sat)}$ = 0.2 V. Which one of the following statements is correct?



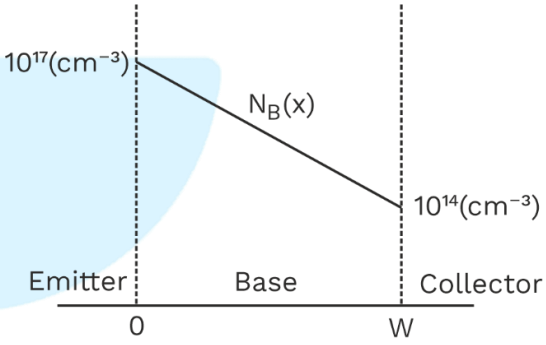
A. For R_C = 1 kΩ, the BJT operates in the saturation region

B. For R_C = 3 kΩ, the BJT operates in the saturation region

C. For R_C =20 kΩ, the BJT operates in the cut-off region

D. For R_C =20 kΩ, the BJT operates in the linear region

Q.6 The base of a npn BJT T1 has a linear doping profile $N_B(x)$ as shown below. The base of another npn BJT T2 has a uniform doping N_B of $10^{17}cm^{-3}$. All other parameters are identical for both the devices. Assuming that the hole density profile is the same as that of doping, the common-emitter current gain of T2 is



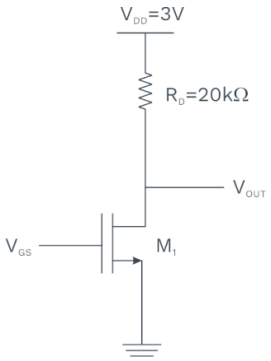
(A) approximately 2.0 times that of T1.

(B) approximately 0.3 times that of T1.

(C) approximately 0.5 times that of T1.

(D) approximately 0.7 times that of T1.

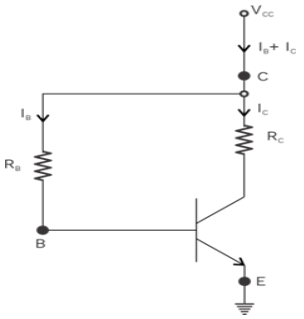
Q.7 For the transistor M_1 in the circuit shown in the figure, $\mu_n C_{ox}$= 100 μA/V^2 and (W/L) = 10, where μ_n is the mobility of electron, C_{ox} is the oxide capacitance per unit area, W is the width and L is the length.



The channel length modulation coefficient is ignored. If the gate-to-source voltage V_{GS} is 1V to keep the transistor at the edge of saturation, then the threshold voltage of the transistor (rounded off to one decimal place) is _____ V.

Q.8 A BJT is connected in the common emitter configuration. If the transistor is operating in the active region, having a current gain α = 0.95, the collector to base reverse bias saturation current = 0.8 µA and the base current I_B = 25 µA, then the collector current of the transistor will be _____ µA. (Rounded off to the nearest integer)

Q.9 The Si based transistor shown in the below figure is biased for constant base current. If V_{BE} = 0.7 V, β = 80, V_{CE} = 9V, R_C = 3 kΩ and V_{CC} = 15 V, th



(Assume that the leakage current and base width modulation are negligible)

(A)Base current = 25 µA

(B)Base Resistance (R_B) = 572 kΩ

(C)Base current = 32 µA

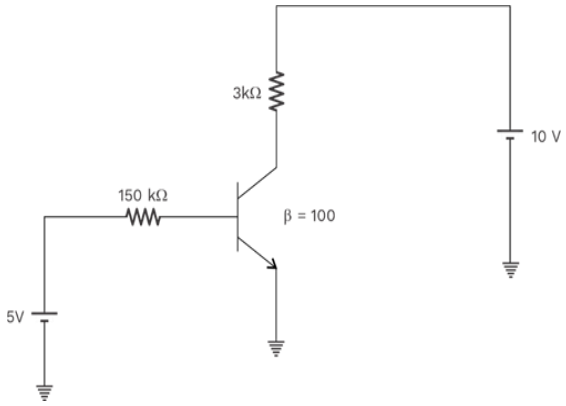
(D)Base Resistance (R_B) = 496 kΩ

Q.10 A PNP transistor is operated in the active region with an emitter injection efficiency of 0.98. Out of 99 mA of the hole current injected into the base, 1.98 mA recombines in the base. The current gain in the common base configuration for the BJT is-

(A)0.3900 (B)0.9604

(C)0.0196 (D)0.0298

Q.11 In the circuit shown below, the transistor has the common-emitter current gain of 100, $V_{BE(ON)}$ = 0.7 V, and $V_{CE(sat)}$ = 0.2 V.



The transistor is biased in-

(A)Saturation Region

(B)Active region

(C)Cut-off Region

(D)Inverse active region

Q.12 For a transistor having $I_{CO} = I_{CBO} = 4$ μA and $\alpha = 0.98$, the value of I_{CEO} = _____ μA.

(Rounded off to the nearest integer)

Q.13 In a silicon BJT at 300K(assuming $\eta = 2$ and $V_T = 0.025$ V), if the emitter current is doubled, the voltage across the base-emitter junction

(A)Increases by 35 mV

(B)gets doubled

(C)gets halved

(D)remains constant

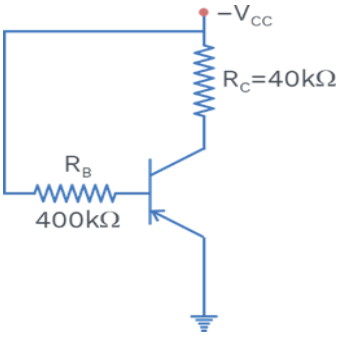
Q.14 A bipolar junction transistor has I_{CBO} = 0.7 mA and a common-base current gain of 0.97. The emitter current is given as 2.8 mA. The transconductance of the transistor is-

(Consider V_T = 26 mV and $\eta = 1$)

(A)131.38 ℧ (B)131.38 m℧

(C)104.46 m℧ (D)104.46 ℧

Q.15 A BJT circuit is shown in the figure below.



For the given PNP transistor:

β = 60

V_{CC} = 12V

I_{CBO} = 16 μA

$|V_{BE}|$ = 0.7 V

The value of collector current I_{CQ} is _____ mA. (Up to 3 decimal places)

Q.16 A p-n-p BJT has emitter doping $(N_E) = 10^{20}/cm^3$, base doping $(N_B) = 10^{18}/cm^3$ and collector doping $(N_C) = 10^{17}/cm^3$. The width of the base is 0.6 μm and collector to base voltage V_{CB} = 60V. Which of the following statements is/are true?

(Consider $\varepsilon_r = 11.8, \quad \varepsilon_0 = 8.85 \times 10^{-14}$ F/cm)

(A)The junction capacitance is 0.121 mF/cm^2

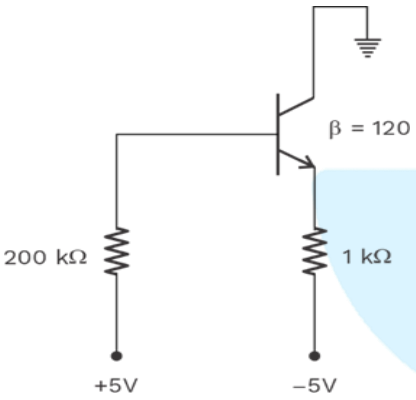
(B)The junction capacitance is 11.25 nF/cm^2

(C)The neutral base width narrowing (ignoring the EB depletion region) is 0.084 μm.

(D)The neutral base width narrowing (ignoring the EB depletion region) is 8.42μ m.

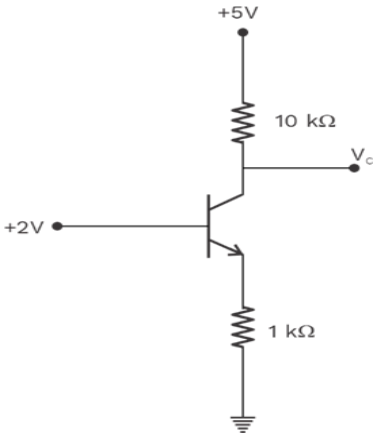
Q.17 Consider the npn transistor circuit shown in the figure below.

Assume, V_{BE} = 0.7 V



The value of collector to emitter voltage V_{CE} is _____V. (Up to one decimal place)

Q.18 For the transistor circuit shown below, β is very large and V_{BE} = 0.7 V



The value of collector voltage V_C is-

(A)-8V (B)2.6V

(C)1.5V (D)None of these

Q.19 A common base transistor has reverse saturation current I_{CBO} = 1μA at 25°C . If this reverse saturation current doubles every 8°C rise in temperature, then find the base current at 57°C.

(Consider α = 0.95, and I_E = 1 mA at 57°C)

(A)34 mA (B)50 mA

(C)34 μA (D)50 μA

Q.20 A transistor in common base configuration has α = 0.99 and leakage current I_{CBO} = 5 μA. If the base current I_B = 100 μA, then the value of emitter current I_E = _____ mA. (Up to one decimal place)

Q.21 For a transistor with common base, the dc current gain is 0.99 and the base current is 20 µA. Choose the correct option(s).

(A)$I_C = 1.98$ mA

(B)$I_C = 2\mu A$

(C)$I_E = 1.98\mu A$

(D)$I_E = 2$ mA

Q.22 The common-mode collector gain (γ) of BJT is 15. Now if the common-base current gain (α) changes 0.5% from its nominal value, then the percentage change in the common-emitter current gain (β) will be-

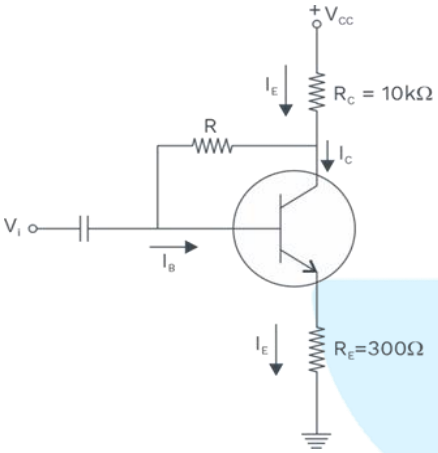
(A)5% (B)3%

(C)2.5% (D)7.5%

Q.23 For the circuit shown below. Find the value of R.

Given,

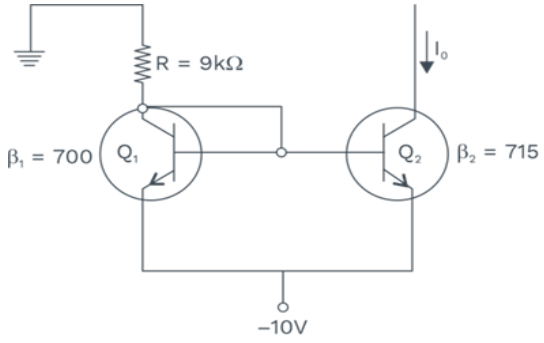
V_{CC} = 30V, R_C = 10KΩ, and R_E = 300Ω, and a silicon transistor is used with β = 50 and V_{CE} = 5V. Neglect the reverse saturation current.



(A)$107.2\text{k}\Omega$ (B)$50\text{k}\Omega$

(C)$90.34\text{k}\Omega$ (D)$30.23\text{k}\Omega$

Q.24 For the silicon Bipolar junction transistor shown below, assume that the emitter area of transistor Q_1 is half that of transistor Q_2. The value of the current I_0 is _____ mA.



Q.25 In a PNP transistor, the width of the base region is 25 μm. The diffusion constant of holes in the base region is 20 cm^2/sec. The average time taken by the injected carriers to travel from emitter junction to collector junction is

(A)15.265 nsec

(B)75.25 nsec

(C)125.32 nsec

(D)156.25 nsec

Q.26 The reverse leakage current of the transistor when connected in CB configuration is 0.2 μA, and when it is connected in CE configuration, reverse saturation current is 20 μA. The value of $\alpha + \beta + \gamma$ is _____.(up to 2 decimal place)

Q.27 A BJT can act as a switch, when it changes from

(A)Cut off to active region

(B)Active to saturation region

(C)Forward active mode to reverse active mode

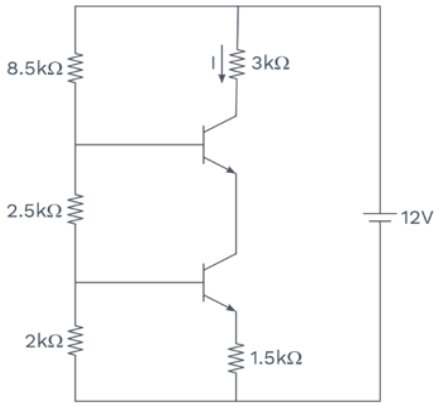
(D)Saturation to cut-off region

Q.28 A bipolar transistor has the following significant parameters at T = 300K.

$\alpha_F = 0.98, \quad \alpha_R = 0.18, \quad I_C = 2\ \text{mA}, \quad I_B = 60\mu\text{A}$

The collector-emitter voltage of saturation is _____ volts.

Q.29 A circuit is shown in the figure below.



β for each transistor is very high and V_{BE} = 0.7V. The current flowing through 3kΩ resistor is _____ mA (up to 3 decimal places)

Q.30 If base width in a bipolar junction transistor is increased, which one of the following statements is correct?

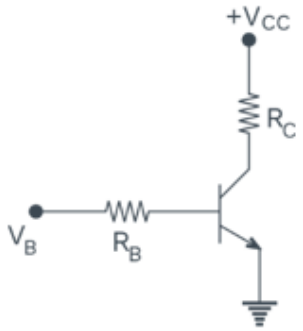
(A)α of the bipolar transistor increases

(B)α of the bipolar transistor decreases

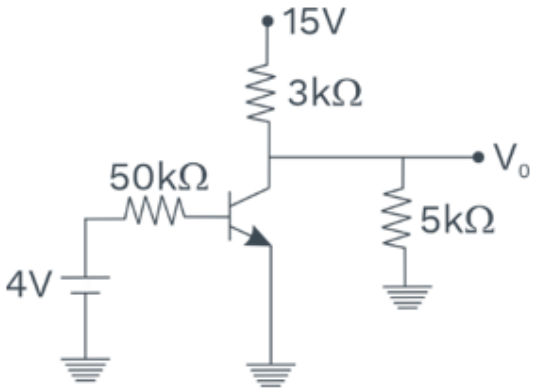
(C)current increases

(D)none of these

Q.31 Consider a silicon based npn transistor in active region as given in the figure below. If the base voltage is 1.7 V and the base resistance is 20 $\mathrm{k}\Omega$, then the base current is _____ mA.



Q.32 An NPN transistor circuit shown below is having a current gain of 150. Which of the following option(s) is/are correct?



(A)The transistor circuit is working in the active region.

(B)The transistor circuit is working in the saturation region.

(C)The collector current I_C is 9.9 mA

(D)V_{CE} voltage is 0.2 V

Q.33 Which of the following is/are correct about “Early Effect” in a BJT?

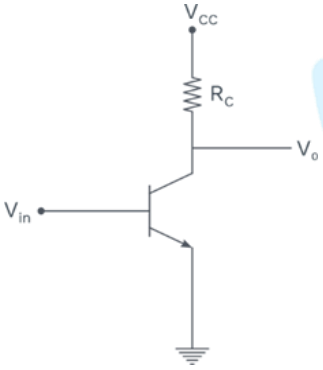
(A)Early effect refers to reduction of effective base width.

(B)It is caused by early removal of stored charge in Base region during saturation to cut-off switching.

(C)It is caused by the reverse biasing of base-collector junction.

(D)Early effect decreases the forward common-emitter current gain

Q.34 The transistor shown in figure below is biased with collector current of 2.6 mA and R_C = 2.5 kΩ . If β = 99 and V_A = 13V, then small signal voltage gain is-



(A)–250 (B)–500

(C)–200 (D)–166.67

Q.35 Which of the following represent the base transport factor for a n-p-n BJT?

(A)$\frac{I_{PC}}{I_{PE}}$ (B)$\frac{I_{NC}}{I_{NE}}$

(C)$\frac{I_{PE}}{I_{PE}+I_{NE}}$ (D)$\frac{I_c}{I_E}$

Q.36 Two transistors T_1 and T_2 has early voltage infinite and 20 V respectively. Both transistors are biased in active region with V_{CE} = 2 volt. To get same collector current base to emitter voltage of

(A)Transistor T_1 is greater than T_2(B)Transistor T_1 is less than T_2

(C)Transistor T_1 is equal to T_2

(D)B and C

Q.37 Which of the following option(s) is/are correct regarding the configuration of BJT?

(A)The common emitter amplifier configuration provides highest power gain among all the configurations.

(B)Common base configuration has lowest input impedance among all the configurations.

(C)Common collector configuration provides voltage gain greater than unity.

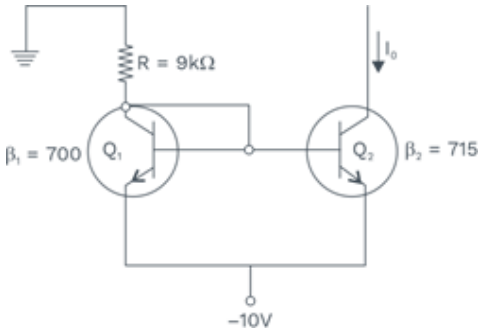
(D)Common collector configuration provides very large current gain.

Q.38 The common-mode collector gain Gamma of BJT is 15. Now if the common base current gain (α) changes 0.5% from its nominal value, then the percentage change in the common-emitter current gain (β) will be-

(A)5% (B)3%

(C)2.5% (D)7.5%

Q.39 For the silicon Bipolar junction transistor shown below, assume that the emitter area of transistor Q_1 is half that of transistor Q_2. The value of the current I_0 is _____ mA.



Q.40 A BJT transistor has current gain of 100 having forward biased junction capacitance of 5pF and reverse biased junction capacitance of 1pF. The β cut-off frequency(in MHz), if $r_\pi = 10\text{k}\Omega$, is

(A)109 (B)2.65

(C)20.9 (D)25.9

Q.41 The NPN transistor made of silicon has a DC base bias voltage of 15V and an input base resistor of 150kΩ. Then the value of the base current into the transistor is-

(A)50.2 µAmp

(B)13.6 µAmp

(C)95.3 µAmp

(D)8.2 µAmp

Q.42 A npn transistor in forward-active mode of operation is based at I_C = 20mA. The total emitter-base capacitance is 200pF and the base transit time is 200psec. The depletion capacitance of the emitter-base junction in pF is _____.

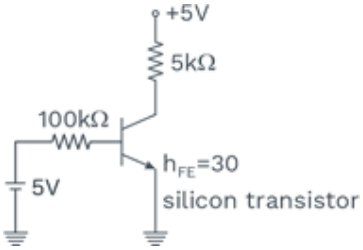
(A) 60Pf

(B)46pF

(C)154pF

(D)55pF

Q.43 The transistor in the circuit of the given figure is operating-



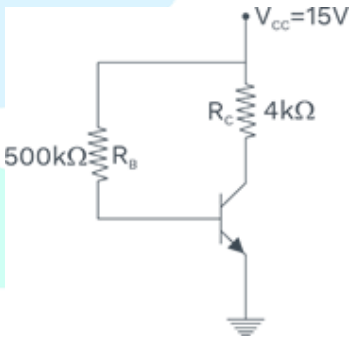
(A)in the cut-off region

(B)in the active region

(C)in the saturation region

(D)either in the active or the saturation region

Q.44 A transistor circuit is shown in figure



Given (V_{CEsat} = 0.2V, β = 100)

The transistor is operating in-

(A)Saturation region

(B)Active region

(C)Cut-off region

(D)Reverse active region

Q.45 Consider the following statements about α-cut off frequency of a bipolar junction transistor-

1. α-cut off frequency is given by $(1+\beta)f_\beta$

2. $f_\alpha < f_\beta$ always

3. It Increases with an increase in base width.

4. It Increases with an increase in collector current.

Which of the following statements are correct?

(A)1 and 2 only

(B)1 and 4 only

(C)1, 2 and 4 only

(D)2, 3 and 4 only

Q.46 The DC current gain (β) of a BJT is 40. Assuming that the emitter injection efficiency of 0.99, the base transport factor is

(A)0.985 (B)0.995

(C)0.99 (D)0.975

Q.47 For a BJT, the common base current gain α = 0.99 and the collector base junction reverse bias saturation current $\mathrm{I_{CO}} = 0.5\mu\mathrm{A}.$ if base current $\mathrm{I_B} = 30\mu\mathrm{A}$. Then collector current $\mathrm{I_c}$ is

(A)2.97 mA (B)3.02 mA

(C)$2.97\mu\mathrm{A}$ (D)$3.02\mu\mathrm{A}$

Answer Key

Q.1 Ans. [A]
Q.2 Ans. [A]
Q.3 Ans. [B]
Q.4 Ans. [5.785, Range is 5.7 to 5.9]
Q.5 Ans. [B]
Q.6 Ans. MTA [Marks to All]
Q.7 Ans. [0.5, Range 0.45 to 0.55]
Q.8 Ans. [491, Range 489 to 493]
Q.9 Ans. [A, B]
Q.10 Ans. [B]
Q.11 Ans. [B]
Q.12 Ans. [200, Range 198 to 202]
Q.13 Ans. [A]
Q.14 Ans. [B]
Q.15 Ans. [2.671, Range 2.6 to 2.8]
Q.16 Ans. [B, C]
Q.17 Ans. [1.5, Range 1.45 to 1.55]
Q.18 Ans. [C]
Q.19 Ans. [C]
Q.20 Ans. [10.5, Range 10 to 11]
Q.21 Ans. [A, D]
Q.22 Ans. [D]
Q.23 Ans. [C]
Q.24 Ans. [2.102, Range 2 to 2.2]
Q.25 Ans. [D]
Q.26 Ans. [199.99, Range 195 to 204]
Q.27 Ans. [D]
Q.28 Ans. [0.16, Range 0.13 to 0.19]
Q.29 Ans. [0.764, Range 0.75 to 0.8]
Q.30 Ans. [B]
Q.31 Ans. [0.05, Range 0.02 to 0.08]
Q.32 Ans. [B, D]
Q.33 Ans. [A, C]
Q.34 Ans. [D]
Q.35 Ans. [B]
Q.36 Ans. [A]
Q.37 Ans. [A, B, D]
Q.38 Ans. [D]
Q.39 Ans. [2.102, Range 2 to 2.15]
Q.40 Ans. [B]
Q.41 Ans. [C]
Q.42 Ans. [B]
Q.43 Ans. [C]
Q.44 Ans. [B]
Q.45 Ans. [B]
Q.46 Ans. [A]
Q.47 Ans. [B]

4 MOSFETs

MCQ, MSQ and NAT

Q.1 If the transconductance of the MOSFET increases in the saturation mode, then it is due to

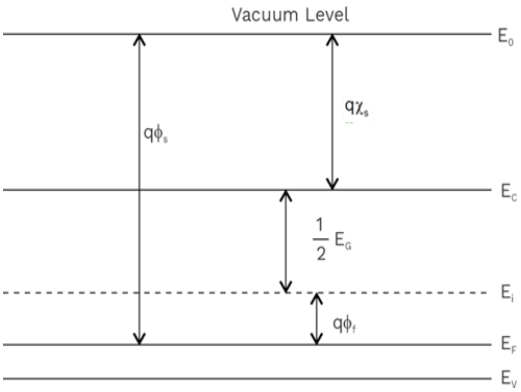
(A)a decrease in the drain to source voltage

(B)a decrease in the channel length

(C)an increase in the drain to source voltage

(D)an increase in the oxide capacitance

Q.2 The energy band diagram of a MOS diode is given below.



Given,

Metal semiconductor work function (ϕ_{ms}) = - 0.792 V

Work function of metal (ϕ_m) = 4.6 V

Electron affinity of the semiconductor (χ_s) = 2.75 V

Acceptor concentration in the semiconductor (N_A) = 10^{14} cm^{-3}

Threshold voltage (V_T) = 0.0259 V

Intrinsic concentration (n_i) = 1.5×10^{10} cm^{-3}

The energy band gap of the semiconductor is_____eV. (Up to two decimal places).

Q.3 The effective width of the silicon bar for p-channel silicon FET is 2×10^{-4} cm and the channel resistivity is 10 Ω-cm. The pinch-off voltage of the FET is-

(Assume μ_p = 500 cm^2/V-sec, μ_n = 1500 cm^2/V-sec and ε_r = 12)

(A)1.92 Volts (B)3.78 Volts

(C)6.52 Volts (D)4.39 Volts

Q.4 A FET has the following parameters.

Saturation value of drain current when gate is shorted to source = 1.75 mA

Pinch-off voltage = -3 Volt

Drain resistance = 5 kΩ

Gate to source voltage = - 3.63 volts

Which of the following parameter(s) of the FET is/are correct?

(A)Drain current, I_D = 0.771 mA

(B)Transconductance, g_m = 0.478 mA/V

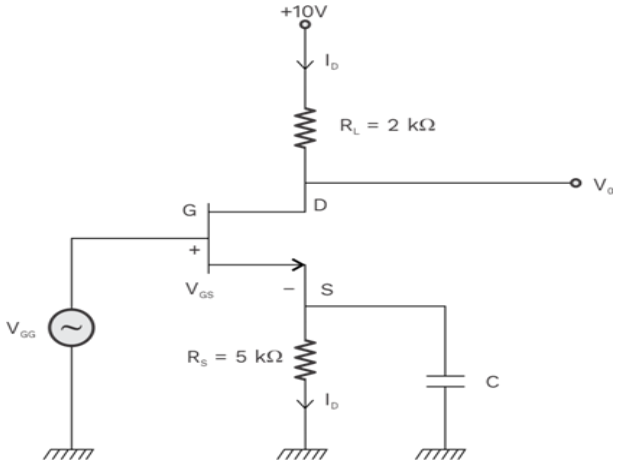
(C)Voltage gain, A_V = 1.21

(D)Voltage gain, A_V = 2.49

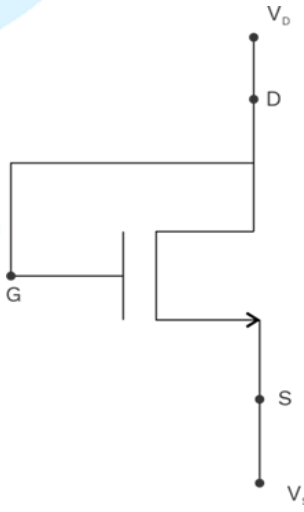
Q.5 An nMOS transistor has $\mu_n C_{ox}$ = 20 μA/V², $\left(\frac{W}{L}\right) = 5$, gate to source voltage = 3V and threshold voltage = 1 V. If the transistor is operating in linear region, then the output conductance of the transistor is_____ μA/V. (Rounded off to the nearest integer)

Q.6 An n-channel MOSFET is biased such that gate to source voltage is equal to drain to source voltage. The threshold voltage (V_T) is 2V. If the drain current is 3.5 mA for V_{GS} = 6V, then for V_{GS} = 8 V, the drain current I_D will be _____mA. (Up to three decimal places).

Q.7 A FET circuit is given below. If I_{DSS} = 5 mA, pinch-off voltage (V_P) = -5V, then drain to source voltage (V_{DS}) of the FET is _____ Volts. (Up to two decimal places)



Q.8 Consider that the N-channel MOSFET shown in the figure is in saturation mode. When drain voltage is varied from 5 to 1 Volt, the current I_D changes from 2 mA to 1 mA. The channel length modulation parameter(λ) is _____V^{-1}. (Up to two decimal places)



Q.9 In a MOS capacitor, the gate voltage exceeds the threshold voltage by 2V. If the oxide thickness is 50 nm, then the inversion charge density of the channel is_____× 10^{-8} C/cm². (Up to one decimal place).

(Assume $\varepsilon_{ox} = 3.9\varepsilon_o$, $\varepsilon_o = 8.85 \times 10^{-14}$ F/cm).

Q.10 Consider an n-channel MOS structure at T = 300 K. If the surface potential is 0.518 V, then the concentration of n-type gate is _____× 10^{13} cm^{-3}. (Up to one decimal place)

(Assume $n_i = 1.5 \times 10^{10}$ cm^{-3}, V_T = 25.9 mV).

Q.11 Consider a MOSFET operating in the saturation region and having a draincurrent of 2.5 mA. If the transconductance (g_m) is 1.7 mA/V, then the overdrive voltage of the MOSFET is_____ (Up to 2 decimal places)

Q.12 Consider a MOS structure that has given specifications.

The circular cross-section having diameter 0.6 mm

SiO_2 layer of thickness 90 nm

The permittivity of SiO_2 is 3

$\varepsilon_0 = 8.854 \times 10^{-14}$ F/cm.

The capacitance of the MOS structure is -

(A)8.98 pF

(B)26.32 pF

(C)1.39 nF

(D)83.22 pF

Q.13 In a particular application, an n channel MOSFET operates with V_{SB} in the range of 0 to 5 V. If V_{tno} (threshold voltage for V_{SB}=0) is 1 V, bulk threshold $(\gamma) = 0.5\sqrt{V}$, and ϕ_f = 0.35, then the threshold voltage is -

(A)0 V to 1.8 V

(B)1 V to 1.77 V

(C)1 V to 2.11 V

(D)0 V to 0.7 V

Q.14 A MOSFET carries a drain current of 1 mA with V_{DS} = 0.5 V in saturation. If V_{DS} rises to 1 V and λ = 0.1 V^{-1}, then which of the following statement(s) is/are true?

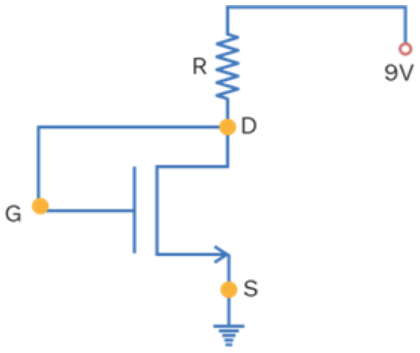
(A)The current I_{D_2} = 1.048 mA

(B)The current I_{D_2} = 48 µA

(C)The output resistance, r_0 = 10.42 kΩ

(D)The output resistance cannot be determined

Q.15 The NMOS circuit shown below has V_T = 1.2 V, $\mu_n C_{ox}$ = 25 µA/V^2, L = 1.5 µm, and λ = 0. If I_{DS} = 0.3 mA and V_{DS} = 3.6 V, then the width of the channel (W) and resistance (R) are respectively.



(A)1.29 µm and 26 kΩ

(B)6.25 µm and 18 kΩ

(C)1.29 µm and 18 kΩ

(D)6.25 µm and 26 kΩ

Q.16 The mobility of the carrier in the channel of a MOSFET is lower than the mobility of the carrier in the bulk semiconductor because-

(A)carriers are scattered by fixed charges in the oxide layer.

(B)carriers in the channel are very close to semiconductor-oxide interface.

(C)higher gate biasing

(D)lower gate biasing.

Q.17 In a MOSFET with sufficient drain voltage in saturation mode (under constant mobility conditions), the current is 4 mA at V_{GS} = 2 V and 24 mA at V_{GS} = 4 V. The threshold voltage (V_T) of the MOSFET is _____ V. (Up to two decimal places)

Q.18 An N-channel MOSFET is operating in the saturation region with drain current (I_D) = 5 mA and drain to source voltage (V_{DS}) = 3 V. If the channel length modulation parameter is $\lambda = 0.05\ V^{-1}$, then the drain to source channel conductance is _____ $\times 10^{-5}$ S (Up to two decimal places)

Q.19 A MOS capacitor is made using an n-type substrate which is in accumulation mode. The dominant charge in the channel is due to the presence of –

(A)electrons

(B)holes

(C)electron-hole pairs

(D)negativity charged ions

Q.20 Consider an NMOS device with parameter ϕ_s = 0.64 V. Find the maximum value of γ such that the shift in threshold voltage between V_{SB} = 0 V to V_{SB} = 10 V is not more than 1.6 V.

(A)0.3499 $V^{1/2}$

(B)0.504 $V^{1/2}$

(C)0.904 $V^{1/2}$

(D)0.6499 $V^{1/2}$

Q.21 An NMOS transistor is operating in the saturation region with output current I_D = 250 μA, $\frac{W}{L} = 10, \mu_n C_{0x} = 289\mu A/V^2$. The minimum voltage between drain and source is –

(A)0.173 V

(B)0.232 V

(C)0.415 V

(D)0.589 V

Q.22 An n-channel enhancement MOSFET is operating in the saturation region. The threshold voltage is 1 V, gate oxide capacitance is 5×10^{-8} F/cm^2, and mobility of electrons 1300 cm^2/V-sec. The width of the channel is 25 μm and the channel length modulation parameter, λ = 0. If the drain current is 0.5 mA for V_{GS} =2V, then the length of the channel (in m) is -

(A)1.625×10^{-6}

(B)1.230 × 10^{-6}

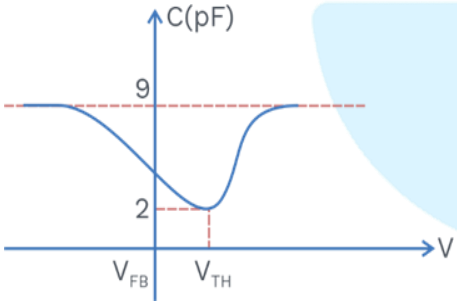
(C)9.01 × 10^{-5}

(D)6.213 × 10^{-7}

Q.23 A CMOS inverter is loaded by 16 pF capacitance. The average current drawn (in mA and up to 3 decimal places) from the voltage source by the CMOS inverter when the inverter switches at a frequency of 1.7 MHz is_____.

(Consider V_{DD} = 12V)

Q.24 The C-V characteristics of a MOS capacitor is shown below.



If $\epsilon_{ox} = 0.32 \times 10^{-12}\ \mathrm{F/cm}; \epsilon_{si} = 1.04 \times 10^{-12}\ \mathrm{F/cm}$ and the thickness of the oxide layer is 1.6 μm, the maximum value of the width of the depletion region is _____ μm. (Up to one decimal place)

Q.25 Consider a p-type silicon substrate at T = 300 K doped with $N_A = 10^{14}$ /cm^3. Let the oxide charge (Q_{ss}') is 10^{10}/cm^2 and the thickness of oxide is 500 Å. The metal semiconductor work function difference ϕ_{ms}) is -0.83V.

(Consider $\varepsilon_r(SiO_2) = 3.9, \varepsilon_r(Si) = 11.7, \varepsilon_0 = 8.85 \times 10^{-14}\ \mathrm{F/cm}$, V_T = 0.0259 V, $n_i = 1.5 \times 10^{10}/cm^3$)

Which of the following statement(s) is/are true?

(A)The maximum space charge width is 2.43 μm

(B)The threshold voltage is -0.397 V

(C)The maximum space charge width is 5.9 μm

(D)The threshold voltage is -3.058 V

Q.26 An n-channel MOSFET has an inversion carrier mobility μ_n = 525 cm^2/V sec. Threshold voltage V_T = 0.75 V and oxide thickness $t_{ox} = 800\,\mathring{A}$ .When MOSFET is in the saturation region, the drain current $I_{D(sat)}$ = 8 mA and V_{GS} = 4 V. The required value of W/L ratio is_____. (Rounded off to the nearest integer)

Assume dielectric constant for SiO_2, $\varepsilon_{rox} = 3.9$.

Q.27 A MOS capacitor has an oxide thickness of $t_{ox} = 400\,\mathring{A}$ and oxide charge density $Q'_{s\ s} = 4 \times 10^{-8}\mathrm{C/cm^2}$. The flat band voltage of the capacitor is-

[Assume work function difference, ϕ_{ms} = -0.898 V, dielectric constant for SiO_2 i.e., ε_{rox} = 3.9]

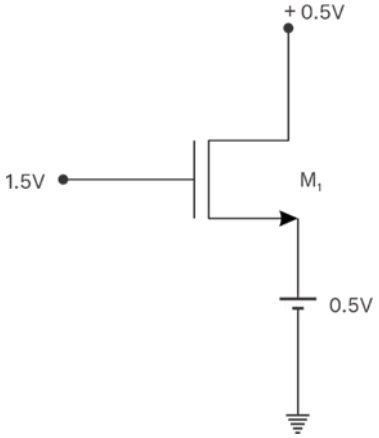
(A)-1.362V

(B)-0.898V

(C)+1.362V

(D)+0.898V

Q.28 A MOSFET circuit is shown below.



If the threshold voltage V_{TH} = 0.4 V, then the region of operation of MOS transistor M_1 is-

(A)Cut-off region

(B)Linear Region

(C)Saturation Region

(D)Cannot be determined

Q.29 Which of the following statement(s) is/are **incorrect** regarding the threshold voltage in MOSFET?

(A)Threshold voltage decreases if the gate oxide thickness increases.

(B)Threshold voltage decrease as the source to body voltage decreases.

(C)Threshold voltage decreases as the substrate doping concentration increases.

(D)Threshold voltage is positive for PMOS and negative for NMOS.

Q.30 Consider an N channel MOSFET in saturation.

Given,

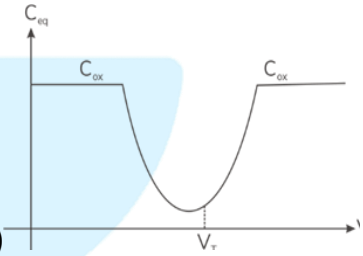
I_D = 110 µA when V_{GS} = 2 V and V_{DS} = 3 V

I_D = 120 µA when V_{GS} = 2 V and V_{DS} = 5 V

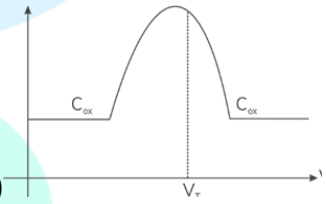
Then the value of λ (channel length parameter) is _____V^{-1}. (up to 3 decimal places)

Q.31 Which one of the following represents a MOS CV plot in a p-type substrate at low frequency?

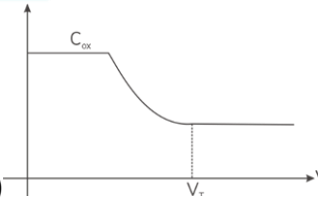
(A)



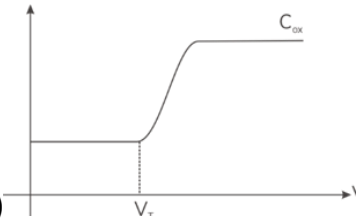
(B)



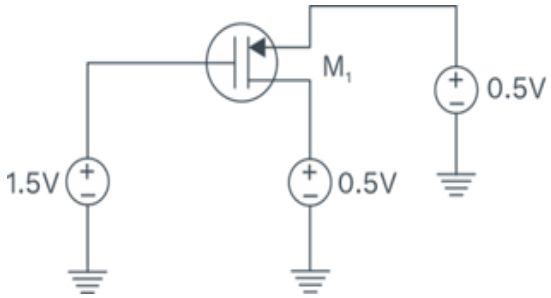
(C)



(D)



Q.32 For the circuit shown below, the region of operation of M_1 is-



[Given, V_T= 0.4 V]

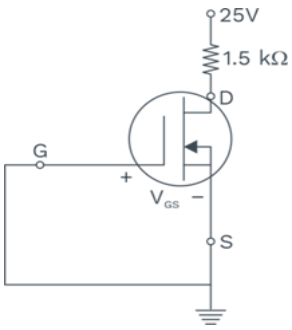
(A)Saturation

(B)Linear

(C)M_1 is off

(D) Cannot be determined

Q.33 For the circuit shown below, the value of V_{DS} is _____ V.



[Given, I_{DSS} = 10 mA, V_P = –4V]

Q.34 Find the pinch-off voltage of a germanium p-channel FET with half channel length of 2 microns and resistivity of 3 Ω-cm . The relative dielectric constant is 16 and the mobility of holes is 1800 cm^2 /sec.

(A)2 (B)2.61

(C)2.9 (D)2.22

Q.35 For an ideal MOS capacitor with n type substate doping concentration of N_D = 2 × $10^{16}/cm^3$. The threshold voltage is 2.1V. If the metal gate of the ideal MOS capacitor is replaced with p^+ polysilicon gate, the threshold voltage of the MOS capacitor becomes _____V (up to 1 decimal place).

Consider,

n_i = 1.5 × $10^{10}/cm^3$

$$\frac{kT}{q} = 26mV$$

Energy gap of S_i = 1.1 eV

Q.36 Consider a MOSFET operating in saturation region having drain current of 2 mA. The overdrive voltage of MOSFET, if transconductance (g_m) is equal to 1.5 mA/V is _____V. (upto 2 decimal places)

Q.37 Which of the following statement(s) is/are correct about threshold voltage of MOSFET?

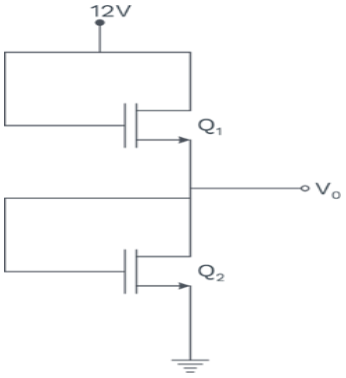
(A)Threshold voltage increases, if gate oxide thickness increases.

(B)Threshold voltage decreases as source to body voltage decreases.

(C)Threshold voltage decreases as substrate doping density increases.

(D)Threshold voltage increases as source to body voltage increases.

Q.38 An Enhancement mode MOSFET is shown in the figure below.



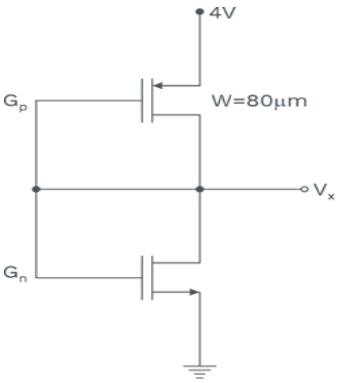
If the transistors are matched, then the output of the circuit is

(Assume Q_1 and Q_2 have same V_T, equal to 1 V)

(A)6V (B)8V

(C)10V (D)12V

Q.39 A MOS transistor circuit is shown in the figure below.



It is given that $\mu_n C_{ox} = 2\mu_p C_{ox} = 20\ \mu A/V^2$, $V_{tn} = |V_{tp}| = 1V$, $\lambda = 0$, $L_n = L_p = 10\ \mu m$ and $W_n = 40\ \mu m$ until it is specified otherwise. The voltage V_x is _____ V.

(Correct to nearest integer value)

Q.40 Choose the correct statement(s) about the threshold voltage of MOSFET.

(A)V_T increases with an increase in oxide thickness.

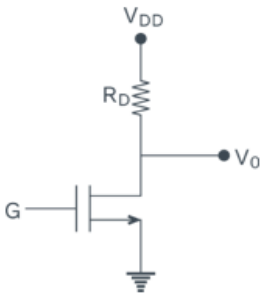
(B)V_T increases with the increase in acceptor concentration.

(C)V_T does not depend upon the oxide thickness.

(D)V_T does not depend upon the impurity concentration.

Q.41 An NMOS transistor is operating in the saturation region. Find the output voltage (V_0), considering the following specifications-

$$I_D = 180\ \mu A, \frac{W}{L} = 10, \mu_n C_{ox} = 390\ \mu A/V^2$$



(A)0.276 (B)0.281

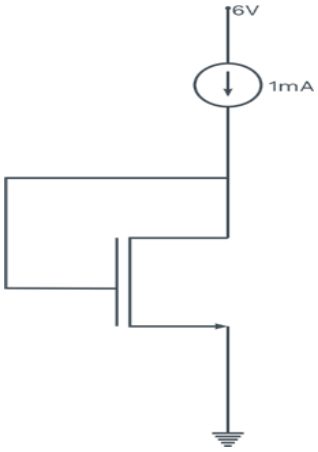
(C)0.30 (D)0.413

Q.42 A MOSFET has a drain current of 1.5mA when V_{GS} = 5V and V_{DS} = 8V with the threshold voltage of 2V. The drain current (in mA), when V_{GS} = 4V and V_{DS} = 8V is_____.

Q.43 Consider the transistor in the circuit having following parameters:

V_{TN} = 1 V, K_n = 1 mAV^{-2} and $\lambda = 1V^{-1}$. The value of voltage V_{DS} is _____V.

(Up to 2 decimal places).



Q.44 A voltage V is applied to a MOS capacitor with P type silicon substrate at T = 300K. The inversion carrier density for V = 1V is $8x10^{10}$ cm^{-2}. For V = 1.4V the inversion carrier density is 2×10^{11} cm^{-2}. Calculate the value of the inversion carrier density for V = 1.8 V.

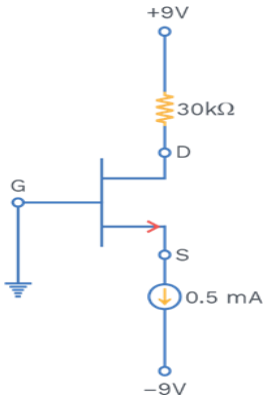
(A)3.6×10^{11} cm^{-2}

(B)3.2×10^{11} cm^{-2}

(C)4.8×10^{11} cm^{-2}

(D)5.6×10^{11} cm^{-2}

Q.45 Parameters of the transistor shown in the figure are V_{TN} = 0.8V and K_n = 0.2 mA/V^2. The voltage V_S is



(A)–2.38 V

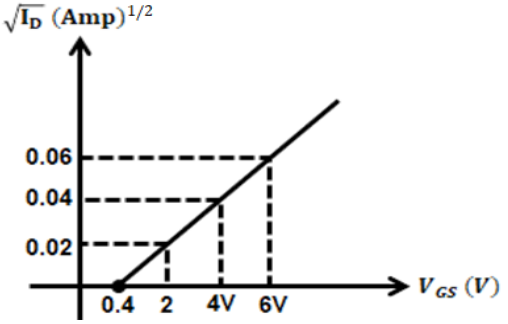
(B)1.32 V

(C)1.72 V

(D)None of these

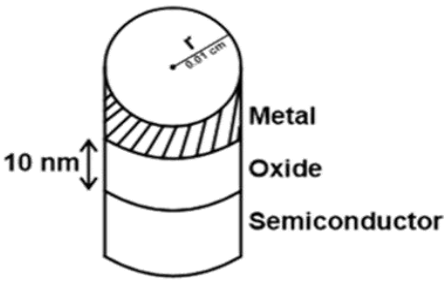
Q.46 In a MOS capacitor oxide breakdown occur when electric field across oxide layer E_{OX} is 10^5 kV/cm. If oxide thickness is 50nm then oxide breakdown voltage is _____ volt.

Q.47 The experimental characteristics of an ideal N-channel MOSFET biased in the saturation region is shown in figure below



If $\frac{W}{L} = 10$, $\varepsilon_{OX} = 3.9\varepsilon_O$, $\varepsilon_O = 8.85 \times 10^{-14}$ F/cm and t_{OX} = 34.5 nm, inversion carrier mobility μn will be _____ $\frac{cm^2}{V - sec.}$

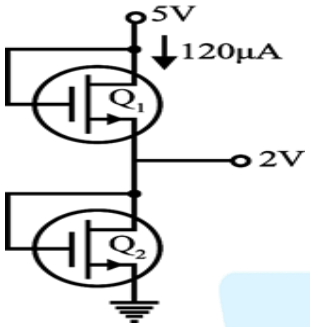
Q.48 Consider a MOS capacitor as shown



Permittivity of oxide layer $\varepsilon_{ox} = 1.2\ \varepsilon_o$ & ε_o = 8.85×10^{-14} F/cm

The oxide capacitance is _____ pF.

Q.49 The NMOS transistors in the circuit shown below have $V_t = 1\ V, \mu_n C_{ox} = 120\mu A/V^2$ and $L_1 = L_2 = 1\mu m$. Required values of gate channel width for each of Q_1, Q_2 to obtain the voltage and current values indicated.



(A)Gate channel width $W_1 = 2\mu m$

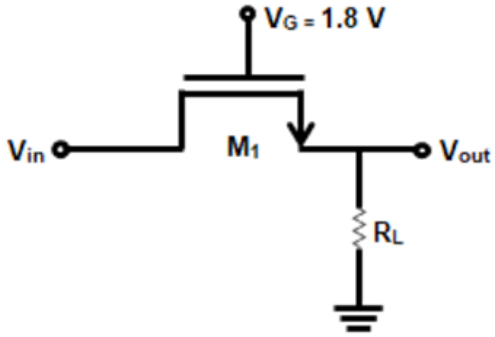
(B)Gate channel width $W_2 = 2\mu m$

(C)Gate channel width $W_1 = 0.5\mu m$

(D)Gate channel width $W_2 = 0.5\mu m$

Q.50 An MOS structure is formed using a p-type silicon wafer with $N_a = 5 \times 10^{16}$ cm^{-3}.The maximum space-charge width is _____ µm. Assume T = 300K and intrinsic carrier concentration, $n_i = 1.5 \times 10^{10}$ cm^{-3} and $\varepsilon_{Si} = 11.7\ \varepsilon_o$.

Q.51 In the circuit M_1 serves as an electronic switch. If $V_{in} \approx 0, \frac{W}{L}$ such that the circuit attenuates the signal by only 5%. Assume V_G = 1.8 V and R_L = 100 Ω is



Given V_T = 0.4 volt

$\mu_n C_{OX}$ = 200 µA/v^2

(A)674 (B)864

(C)764 (D)974

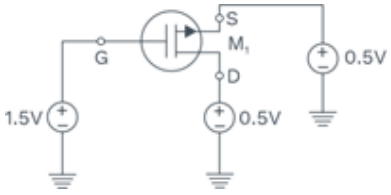
Q.52 A NMOS transistor whose threshold voltage is 0.7V operating in saturation region has drain current of 100µA when gate to source voltage is 1.02V. The value of Gate to source voltage required to cause the device to operate as 1000Ω resistor in triode region is _____ volts.

Q.53 An NMOS transistor has a parameter, K_n of $60\mu A/V^2$ and threshold voltage of 1V at $V_{GS} = 8\ V$ and $I_D = 1\ mA$, then the value of V_{DS} (in volts) is _____.

(Correct up to 2 decimal places)

Q.54 For the circuit shown below, the region of operation of M_1 is-

Given, V_T= 0.4 V



(A)Saturation

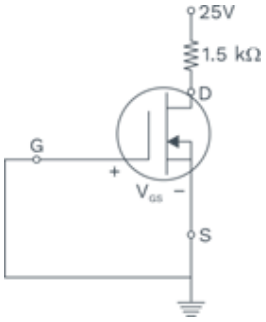
(B)Linear

(C)M_1 is off

(D)Cannot be determined

Q.55 For the circuit shown below, the value of V_{DS} is ..... V.

Given, I_{DSS} = 10 mA, V_P = –4V



Q.56 Find the pinch-off voltage of a germanium p channel FET with half channel length of 2 microns and resistivity of 3 Ω-cm . The relative dielectric constant is 16 and the mobility of holes is 1800 cm^2 /sec.

(A)2 (B)2.61

(C)2.9 (D)2.22

Q.57 Find out the capacitance of a MOS structure with the given specifications.

Cross-sectional diameter = 0.5 mm

Oxide thickness = 50 nm

Relative permittivity of SiO_2 = 4

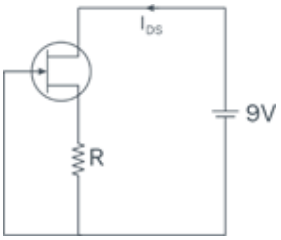
$\epsilon_0 = 8.854 \times 10^{-12}\ \mathrm{F/m}$

(A)196 pF (B)138 pF

(C)126 pF (D)168 pF

Q.58 A MOSFET operating at I_D = 0.1 mA (In Saturation region) has trans conductance (g_m) of 1 mA/V. If $\mu_n C_{OX}$ = 50 $\mu A/V^2$ then $\frac{W}{L}$ ratio is _____.

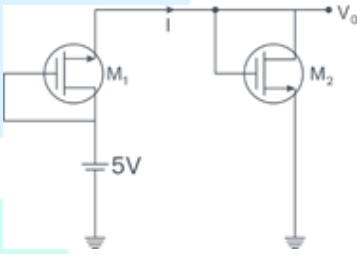
Q.59 The JFET shown below has I_{DSS} = 10mA and V_p = –4V. The value of the resistance R for a drain current I_{DS} = 8.1 mA is



(A)938.27 Ω (B)49.38 Ω

(C)1 kΩ (D)93.82 Ω

Q.60 Consider the circuit shown below:



If both the MOS transistors are identical, with $\frac{\mu_n C_{ox} W}{L} = 80\mu A/V^2$ and $V_T = 2V$, then the value of current I is-

(A)$10\mu A$ (B)$6.5\mu A$

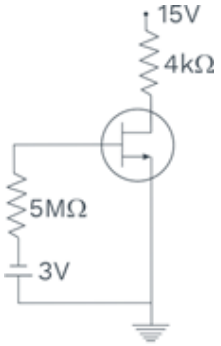
(C)$5\mu A$ (D)$8.2\mu A$

Q.61 A JFET has $g_m = 20\mathrm{mS}, I_{DSS} = 15\mu A.$ Calculate $V_{GS}(OFF)$

(A)–2mV (B)–1.5mV

(C)2V (D) 0.75mV

Q.62 A JFET transistor is shown below. What is the operating point under DC condition if drain saturation current is 6mA and pinch-off voltage is –6V?



(A)I_D = 1.5mA, V_{DS} = 9V

(B)I_D = 3mA, V_{DS} = 4.5V

(C)I_D = 1mA, V_{DS} = 11V

(D)I_D = 1.9mA, V_{DS} = 9V

Q.63 For a certain JFET, I_{DSS} = 10mA, and the cut-off value of gate to source voltage is equal to –6V. What is the drain current for V_{GS} = –1V? (consider that, JFET is operating in saturation region)

(A)6.67 mA

(B)8.33 mA

(C)5 mA

(D)2.5 mA

Q.64 An n channel JFET has I_{DSS} = 4 mA and $V_P = -2V,$ if the gate to source voltage V_{GS} is –1V, then trans conductance g_m for given JFET is-

(A)$2\mu A/V$

(B)2 A/V

(C)0.2 mA/V

(D)2 mA/V

Q.65 For an n-channel JFET, if I_{DSS} = 5 mA and V_p = –20V, then the maximum transconductance is -

(A)$0.2\ m\mho$ (B)$0.4\ m\mho$

(C)$10\ m\mho$ (D)$0.5\ m\mho$

Q.66 Statement-(I): The phenomenon of thermal runaway is not possible at room temperature in JFET.

Statement-(II): The carrier mobility decreases with the increase in the temperature in JFET.

(A)Both Statements (I) and (II) are individually true and Statement (II) is the correct explanation of Statement (I)

(B)Both Statement (I) and (II) are individually true but Statement (II) is not the correct explanation of Statement (I)

(C)Statement (I) is true, but Statement (II) is false

(D)Statement (II) is true, but Statement (I) is false

Q.67 Consider the following statements:

1.BJT is unipolar device.

2. BJT takes more steps to fabricate and takes more space in the IC form.

3. FET takes less space in IC form.

4. FET is a more noisy device.

Which of the above statements are correct?

(A)1, 2, 3 only

(B)2, 3, 4 only

(C)2, 3 only

(D)1, 2, 3 and 4

Answer Key

Q.1 Ans. [B, D]
Q.2 Ans. [4.83, Range 4.8 to4.86]
Q.3 Ans. [B]
Q.4 Ans. [A, C]
Q.5 Ans. [200, Range 198 to 202]
Q.6 Ans. [7.875, Range 7.75 to 9.95]
Q.7 Ans. [5.45, Range 5.4 to 5.5]
Q.8 Ans. [0.33, Range 0.3 to 0.36]
Q.9 Ans. [13.8, Range 13.5 to 14.2]
Q.10 Ans. [33, Range 32.5 to 33.5]
Q.11 Ans. [2.94, Range 2.9 to 3]
Q.12 Ans. [D]
Q.13 Ans. [B]
Q.14 Ans. [A, C]
Q.15 Ans. [B]
Q.16 Ans. [A, B, C]
Q.17 Ans. [0.62, range 0.6 to 0.64]
Q.18 Ans. [21.74, Range 21 to 22]
Q.19 Ans. [A]
Q.20 Ans. [D]
Q.21 Ans. [C]
Q.22 Ans. [A]
Q.23 Ans. [0.326, Range 0.3 to 0.35]
Q.24 Ans. [18.2, Range 18 to 18.5]
Q.25 Ans. [A, B]
Q.26 Ans. [70, Range 69 to 71]
Q.27 Ans. [A]
Q.28 Ans. [B]
Q.29 Ans. [A, C, D]
Q.30 Ans. [0.052, Range 0.04 to 0.06]
Q.31 Ans. [A]
Q.32 Ans. [B]
Q.33 Ans. [10, Range 9.9 to 10.1]
Q.34 Ans. [B]
Q.35 Ans. [-1.2, Range -1.25 to -1.15]
Q.36 Ans. [2.67, Range 2.63 to 2.7]
Q.37 Ans. [A, B, D]
Q.38 Ans. [A]
Q.39 Ans. [2, Range 1.99 to 2.1]
Q.40 Ans. [A, B]
Q.41 Ans. [C]
Q.42 Ans. [0.66, Range 0.6 to 0.7]
Q.43 Ans. [1.62, Range 1.55 to 1.7]
Q.44 Ans. [B]
Q.45 Ans. [A]
Q.46 Ans. [500, Range 496 to 504]
Q.47 Ans. [200, Range 198 to 202]
Q.48 Ans. [33.3, Range 32 to 35]
Q.49 Ans. [B, C]
Q.50 Ans. [0.142, range 0.1 to 0.2]
Q.51 Ans. [A]
Q.52 Ans. [1.212, Range 1.15 to 1.27]
Q.53 Ans. [3.04, Range 2.95 to 3.1]
Q.54 Ans. [B]
Q.55 Ans. [10, Range 9.8 to 10.2]
Q.56 Ans. [B]
Q.57 Ans. [B]
Q.58 Ans. [100, Range 98 to 102]

Q.59 Ans. [B]

Q.60 Ans. [A]

Q.61 Ans. [B]

Q.62 Ans. [A]

Q.63 Ans. [B]

Q.64 Ans. [D]

Q.65 Ans. [D]

Q.66 Ans. [A]

Q.67 Ans. [C]

5 Special Purpose Diodes

MCQ, MSQ and NAT

Q.1 A pn junction solar cell of area 1.0 cm^2, illuminated uniformly with 100 $mWcm^{-2}$, has the following parameters: Efficiency = 15%, open circuit voltage = 0.7 V, fill factor = 0.8 and thickness = 200 µm. The charge of an electron is 1.6×10^{-19} C. The average optical generation rate (in $cm^{-3}s^{-1}$) is

(A) 0.84×10^{19}

(B) 5.57×10^{19}

(C) 1.04×10^{19}

(D) 83.60×10^{19}

Q.2 The quantum efficiency (η) and responsivity (R) at a wavelength λ (in µm) in a p-i-n photo detector are related by:

(A) $R = \frac{\eta \times \lambda}{124}$

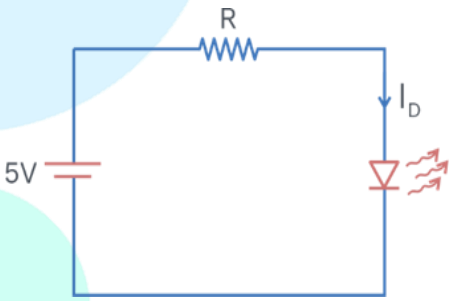
(B) $R = \frac{\lambda}{\eta \times 124}$

(C) $R = \frac{124 \times \lambda}{\eta}$

(D) $R = \frac{124}{\eta \times \lambda}$

Q.3 The saturation velocity of charge carriers of the Avalanche photodiode generated by impact ionization by photon absorption is 2.5×10^9 cm/sec and the width of the depletion region is $100\mu m$. The transition time is _____ psec. (Rounded off to the nearest integer)

Q.4 Consider the circuit shown in the figure below.



The constant forward voltage drop of the LED, i.e., V_D = 2.5 V and the luminous intensity varies directly with the forward current as:

Luminous intensity (I_v) = 30

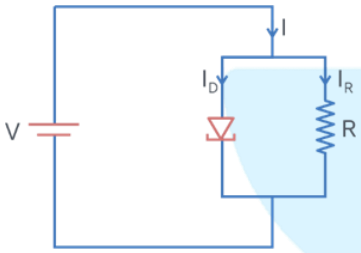
I_D (millicandela)

If the limiting resistance (R) = 100Ω, then the luminous intensity of LED is ____ millicandela(mcd). (Up to 2 decimal places)

Q.5 The reverse saturation current density is $4.1 \times 10^{-11} A/cm^2$ and the

photocurrent density is 19 mA/ cm². The open-circuit voltage of the solar cell is _____V. (Up to 3 decimal places) [Consider that the thermal voltage is 26 mV.]

Q.6 A resistor is placed in parallel with a tunnel diode which has $\left|\frac{dI_D}{dV}\right| = \frac{1}{132\Omega}$. Find the value of the 'R' such that there is no negative resistance region in the V-I characteristics.



(A)$R \leq 7.57\text{k}\Omega$ (B)$R \leqslant 132\Omega$

(C)$R \geqslant 1.32k\Omega$ (D)$R \geqslant 0.132\text{M}\Omega$

Q.7 An n^+p Si junction solar cell has the surface area of $1cm^2$ and acceptor concentration of $N_A = 10^{15}/\text{cm}^3$ in the base region. Assume that the diffusion coefficient of electrons is $40\ cm^2/sec,$ carrier lifetime of electrons is $2.62\mu\text{sec}$, electron-hole pair generation rate is $2.8 \times 10^{19}/\text{cm}^3 - \text{sec}$. The open circuit voltage of the solar cell is _____V. (Up to 3 decimal places)

(Assume V_T = 26 mV, $n_i = 1.5 \times 10^{10}/\text{cm}^3$)

Neglect the contribution of the n^+ region and the depletion region.

Q.8 An InGaAs PIN photodetector has a quantum efficiency of 89% and an energy band gap of 1.45 eV. The number of photons incident on the PIN photodetector is $7x10^6$. Which of the following statement(s) is/are TRUE?

(A)Maximum wavelength = 0.8551 µm

(B)Electron-hole pair generated = $0.623x10^9$

(C)Electron-hole pair generated = $6.23x10^6$

(D)Maximum wavelength = 8.63 Å

Q.9 A tunnel diode is a PN junction diode. Which of the following statement(s) is/are true for a tunnel diode?

(A)In the reverse bias condition, the diode exhibits negative resistance.

(B)Tunnel diode can act as an oscillator.

(C)Fermi level lies on the n-side inside the valence band.

(D)The peak current is decided by the impurity concentration and the junction area.

Q.10 The quantum efficiency of a Si avalanche photodiode is 70 % to detect the radiation of wavelength 0.7 µm. The incident optical power is 0.5 µW. The output current from the device after avalanche gain is 14 µA. The multiplication factor of the photodiode under the given

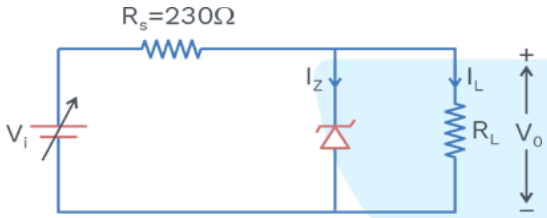
conditions is –

(Consider h = 6.625 × 10^{-34} J-s)

(A)71.065 (B)52330.82

(C)83.92 (D)70987.63

Q.11 An 8 V Zener diode given in the circuit shown below is used as a regulator. If the knee current is negligible, then $r_z = 15\Omega$ and $r_Z << R_L$ (load resistance). If the input voltage (V_i) varies from 15V to 26V, then V_0 varies from 10 V to _____V. (Up to 3 decimal places)



Q.12 Which of the following statement(s) is/are true for the Schottky diodes?

(A) A Schottky diode is also referred to as a hot carrier diode.

(B) The flow of current is controlled by the flow of majority carriers.

(C)It is usually used for rectification of signals for frequencies upto 300 MHz.

(D)Schottky diodes cannot be used in clamping circuits.

Q.13 The incident light power density of the photodiode is 2000 W/m^2 and its area is 6 cm^2. If the efficiency of the diode is 17 %, then the power delivered to the load is _____ mW. (Rounded off to nearest integer)

Q.14 A silicon varactor diode has a capacitance of 60 pF when no bias is applied to the diode. If the reverse bias voltage of 8 V is applied to the diode, then the junction capacitance will be _____ pF. (Up to one decimal place)

[Assume the junction is diffused type and diode barrier potential, V_k = 0.7 V.

Q.15 A photo surface is having a spectral sensitivity of 5 mA/watt and the wavelength of incident radiation is 0.5 μ m. The number of electrons emitted by a pulse of radiation having 11000 photons for given wavelength will be_____. (Rounded off to the nearest integer)

Q.16 Which of the following is/are the application(s) of a varactor diode?

(A)Varactor diode can be used as a switching device

(B)Varactor diode is used in voltage-controlled oscillator circuits.

(C)Varactor diode can be used as a frequency tuner.

(D)Varactor diode is used in low noise amplifiers.

Q.17 Which of the following statements(s) is/are **correct** about light emitting diode (LED)?

(A)The efficiency of a LED decreases with a decrease in the injected current.

(B)The colour of light of a LED depends on the concentration of dopant atoms.

(C)The efficiency of a LED increases with an increase in temperature.

(D)LED does not work on the principle of Electro-luminescence.

Q.18 If the responsivity of the photodiode is 0.6 A/W and the photocurrent generated is 6μA, then the optical power incident on the photodiode is-

(A)3.6 μW

(B)36 μW

(C)1 μW

(D)10 μW

Q.19 A silicon avalanche photo diode has quantum efficiency of 70% at a radiation wavelength λ = 800 nm. If the incident power is 0.5 μW, then the responsivity (R in A/W) is _____. (rounded off upto two decimal places)

Q.20 A photodiode has an efficiency of 75%, when photons of energy 2.5×10^{-19}J are incident upon it. Which of the following statement(s) is/are correct? Consider: Planck constant (h) = 6.625×10^{-34} Js

(A)Responsivity, R = 0.48 A/W

(B)Responsivity, R = 0.64 A/W

(C)Photodiode is operating at wavelength of 7.95 μm

(D)Photodiode is operating at wavelength of 0.795 μm

Q.21 Which of the following statement(s) is/are correct?

(A)Tunnel diode is used in microwave oscillator.

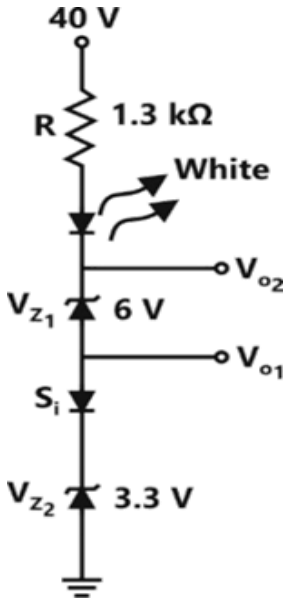
(B)Tunnel diode is used in automatic frequency control circuits.

(C)Varactor diode is used in high speed switch.

(D)Tunnel diode acts as a conductor in reverse bias.

Q.22 A pn junction solar cell illuminated uniformly with 100 Wcm^{-2} has following parameters: Area = $2cm^2$, efficiency = 25%, open circuit voltage = 0.8V, fill factor = 0.8, and thickness = 100μm. The average optical generation rate is _____$\times 10^{22}$ (in cm^{-3} S^{-1}).

Q.23 What is the level of current through the LED [voltage across LED is 4V] and the power delivered by the supply?



(A)40mA, 800mW

(B)20mA, 400mW

(C)20mA, 800mW

(D)40mA, 400mW

Q.24 Which of the following statements is/are true with respect to tunnel diode?

(1) it is high speed device.

(2) it exhibits a negative resistance region.

(3) heavy doping is done only in p side.

(4) peak current I_p> valley current I_V.

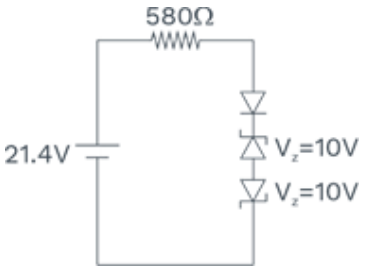
(A)2, 3

(B)1, 2

(C)1, 2, 4

(D)1, 2, 3, 4

Q.25 For the diode in the circuit, given dynamic forward resistance = 5Ω and dynamic reverse break-down resistance = 10Ω. Assuming that all the diodes are made up of silicon material and identical in construction. Then current(I) in the circuit is-



(A)15mA

(B)19.67mA

(C)19.5mA

(D)16.67mA

Q.26 Consider the following statements. Choose the correct option.

(1)Solar cell operates in 3rdquadrant

(2)Solar cell operates in 4thquadrant

(3)Photo-detector with high sensitivity operates in 3rdquadrant

(4)Photo-detector with high sensitivity operates in 1stquadrant

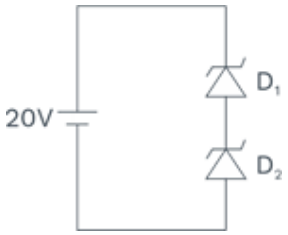
(A)1, 3

(B)2, 4

(C)1, 4

(D)2, 3

Q.27 Consider the circuit diagram-



Which of the following statement is true, if $V_{Z_1} = 10\ \text{V}, V_{z_2} = 15\ \text{V}$ and. $(I_{z\,\min})_{D_1} > (I_{z\,\min})_{D_2}$

(A)Both D_1 and D_2 are in breakdown

(B)Only D_1 is in breakdown

(C)Only D_2 is in breakdown

(D) No diode is in breakdown

Q.28 Which of the following **is not used** for manufacturing an LED?

(A)Silicon oxide

(B)Gallium arsenide

(C)Silicon carbide

(D)Indium gallium arsenide

Q.29 Which of the following statements are correct for light-emitting diodes (LED)?

1. LED uses materials of Gallium and arsenide.
2. LED emits light.
3. LED needs small power for operation.

(A)1 and 2

(B)Only 2

(C)2 and 3

(D)1, 2 and 3

Answer Key

Q.1 Ans. [A]

Q.2 Ans. [A]

Q.3 Ans. [4]

Q.4 Ans. [0.75, Range 0.72 to 0.78]

Q.5 Ans. [0.518, Range 0.48 to 0.53]

Q.6 Ans. [B]

Q.7 Ans. [0.509, Range 0.47 to 0.53]

Q.8 Ans. [A, C]

Q.9 Ans. [B, D]

Q.10 Ans. [A]

Q.11 Ans. [10.673, Range 10.4 to 10.8]

Q.12 Ans. [A, B]

Q.13 Ans. [204, Range 202 to 206]

Q.14 Ans. [21.89, range 21 to 22.3]

Q.15 Ans. [136, range 134 to 138]

Q.16 Ans. [B, C, D]

Q.17 Ans. [A, B]

Q.18 Ans. [D]

Q.19 Ans. [0.45, Range 0.4 to 0.5]

Q.20 Ans. [A, D]

Q.21 Ans. [A, D]

Q.22 Ans. [1.22, Range 1.2 to 1.24]

Q.23 Ans. [C]

Q.24 Ans. [C]

Q.25 Ans. [D]

Q.26 Ans. [D]

Q.27 Ans. [C]

Q.28 Ans. [A]

Q.29 Ans. [D]